



A multiobjective multiperiod mean-semientropy-skewness model for uncertain portfolio selection

Shan Lu¹ · Ning Zhang¹ · Lifan Jia¹

Accepted: 9 October 2020 / Published online: 7 January 2021
© Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract

Due to the complexity of the financial market, security returns are sometimes expressed by expert estimates rather than historical data. In this paper, we deal with a multiobjective multiperiod portfolio selection problem based on uncertainty theory. We propose a new uncertain multiobjective multiperiod mean-semisentropy-skewness portfolio optimization model, in which uncertain semi-entropy is used to quantify the downside risk. To be more realistic, several constraints are also considered, such as the transaction costs, cardinality, liquidity, budget, and bound constraint. Moreover, a novel hybrid technique, called the MFA-SOS algorithm, which combines the features of the firefly algorithm (FA) and symbiotic organism search algorithm (SOS) is designed to solve the proposed model. Finally, a numerical example is given to illustrate the effectiveness of the proposed approach.

Keywords Uncertain portfolio selection · Multi-objective optimization · Semi-entropy · Cardinality constraint · Firefly algorithm

1 Introduction

Portfolio selection involves the optimal allocation of one's capital among different securities. Markowitz [1] first proposed a mean-variance model, in which the investment return and risk are quantified as the expected value and variance, respectively. Later, the mean-variance model was extended by many scholars, such as Sharpe [2] and Atsushi [3]. Moreover, several other risk measures have been introduced, such as the semi-variance [4], absolute deviation [5], value-at-risk (VaR) [6] and conditional value-at-risk (CVaR) [7], etc. In addition, entropy can also be a risk measure and has attracted the attention of many scholars. A study carried out by Philippatos and Wilson [8] was the pioneering work to associate entropy with risk measure in portfolio optimization. Nawrocki and Harding [9] provided two alternative methods to calculate entropy, and entropy was used as the risk measure. Simonelli [10] showed that entropy as a measure of portfolio risk is more convincing than variance. However, a significant

disadvantage of entropy as a risk measure is that it makes no distinction between gains and losses. In other words, it regards high returns, which investors prefer, as equally undesirable as low returns, which investors dislike. The reason is that in addition to low returns, high returns will also contribute to extreme entropy. In particular, when the probability distributions of security returns are asymmetric, entropy becomes an insufficient investment risk measure because it may sacrifice too much expected returns in eliminating both low and high return extremes. To solve this problem, semi-entropy [11] is proposed as a risk measure, which only measures the uncertainty in returns below the expected value and gauges no uncertainty in returns above the expected value. It is more consistent with investors' intuition about risk. Thus, in this paper, semi-entropy is used as a downside risk measure of a portfolio.

In the above studies, the security returns were regarded as random variables. However, due to the complexity of the financial market, enough historical data may not be available. In this situation, some researchers, such as Chen et al. [12], Mansour et al. [13], Zhou and Li [14], introduced fuzzy set theory proposed by Zadeh [15] to study the fuzzy portfolio selection problem. However, some studies [16, 17] have shown that there are paradoxes if fuzzy variables are used to describe security returns. In order to deal with this problem, Liu [16, 18] proposed

✉ Lifan Jia
jialifen@cueb.edu.cn

¹ School of Management and Engineering, Capital University of Economics and Business, Beijing, China

an axiomatic uncertainty theory. Up to now, uncertainty theory has been applied to deal with various optimization problems. Particularly, uncertainty theory has been applied to portfolio optimization. Zhai and Bai. [19] proposed an uncertain mean-risk model with background risk. Chen et al. [20] proposed a new uncertain mean-variance-skewness portfolio selection model with real constraints. Wang and Huang [21] developed an uncertain portfolio optimization model with options by using risk index. Li et al. [22] defined a new uncertain maximin risk measure and derived a new method for constructing the analytic Pareto optimal solution set of the portfolio selection problem. Huang and Yang [23] proposed an uncertain mean-variance model with background risk and explored how background risk affects individual investment decisions. The above literatures focused only on uncertain single-period portfolio selections. However, in the real world, investors often dynamically adjust their investment strategies according to the market environment to obtain the optimal terminal wealth. Hence, the uncertain multi-period portfolio selection problem has attracted the attention of many scholars. For example, Huang and Qiao [24] presented a multi-period uncertain risk index model. Li et al. [25] formulated an uncertain multi-period portfolio selection problem with bankruptcy constraint by a new method in three steps. Chen et al. [26] proposed a new multi-period uncertain mean-variance-skewness model with cardinality constraint. Zhang [27] presented a novel multi-period mean-absolute deviation uncertain portfolio selection model and designed a novel discrete iteration method. Chen et al. [28] proposed a multi-period mean- semivariance portfolio optimization model with real-world constraints.

The limitation on the number of securities is a common consideration for investors. The introduction of cardinality constraint transforms the portfolio selection problem into a nonlinear mixed quadratic programming problem. The traditional exact solution approaches will consume too much computation time cost as the problem scale increases. In such situations, the introduction of metaheuristic algorithm is imperative. The use of three heuristic algorithms based on genetic algorithms, tabu search and simulated annealing for cardinality constrained portfolio optimization were proposed and discussed by Chang et al. [29]. In [26] and [28], two improved imperialist competitive algorithms (ICA), were used to solve the uncertain portfolio selection problems with cardinality constraint. Besides, Deng et al. [30] used improved particle swarm algorithms (PSO) to solve markovitz-based portfolio optimization with cardinality constraint. Note that the metaheuristic algorithm in the aforementioned studies are used for solving single-objective optimization problems. Metaheuristic algorithms also show great advantages in dealing with multi-objective optimization problems (see for example [31–35]).

Particularly, metaheuristic algorithms are widely applied to multi-objective portfolio selection problems. For instance, Wang et al. [36] proposed a fuzzy simulation based on multi-objective particle swarm optimization algorithm for a multi-objective portfolio selection model with fuzzy Value-at-Risk ratio. Chen and Xu [37] developed a hybrid multi-objective bat algorithm for fuzzy portfolio optimization with cardinality constraint. Silva et al. [38] developed a multi-objective particle swarm algorithm with a novel adaptive ranking procedure for mean-variance portfolio selection problem. Kaucic [39] proposed an improved multi-objective particle swarm optimization algorithm and a hybrid constraint-handling approach for portfolio optimization with cardinality constraint. Liagkouras [40] presented a new three-dimensional encoding multi-objective evolutionary algorithm for portfolio optimization.

Recently, FA as an advanced metaheuristic algorithm was proposed by Yang [41] in 2008. This algorithm simulates the behavior of fireflies based on their flash characteristics. Compared with some widely used metaheuristic algorithms, FA has the characteristics of fast convergence speed and simplicity. Thus, FA has been widely used in optimization problems, such as the flow shop scheduling problem [42] and electric power plant planning problem [43]. The multi-objective firefly algorithm (MOFA) was extended by Yang [44] for multi-objective optimization problem. Following this work, Marichelvam et al. [45] introduced a discrete firefly algorithm to solve the multi-objective hybrid flow shop scheduling problem. Karthikeyan et al. [46] proposed a a hybrid discrete firefly algorithm to solve the multi-objective flexible job shop scheduling problem. Hidalgo-Paniagua et al. [47] used the MOFA to solve the multi-objective path planning problem. Bozorg-Haddad et al. [48] used an extended multi-objective developed firefly algorithm for hydropower energy generation. Nguyen et al. [49] used the multi-objective firefly algorithm to deal with the localization problem in wireless sensor network. Lu et al. [50] proposed a hybrid MFA-SCA to solve a multi-objective multi-period regret minimization uncertain portfolio selection model with bankruptcy constraint.

Symbiotic organism search (SOS) is a robust metaheuristic algorithm developed by Cheng and Prayogo in 2014 [51]. SOS has been utilized in the optimization of many problems, such as wireless sensor networks [52] and structural optimization [53]. Moreover, the multi-objective variants of SOS also show good performance in multi-objective optimization. Tran et al. [54] presented a novel opposition multiple objective symbiotic organisms search for solving the time-cost-labor utilization tradeoff problem. Tejani et al. [55] proposed a multiobjective adaptive symbiotic organisms search with a two-archive technique for truss optimization problem. Abdullahi et al. [56] proposed a chaotic symbiotic organisms search algorithm to

solve multi-objective large scale task scheduling optimization problem on IaaS cloud computing environment. Tejani et al. [57] designed a multi-objective modified adaptive symbiotic organism search with two modified phases for solving multi-objective truss design problems.

According to “no-free-lunch” theorem, a metaheuristic algorithm cannot be used appropriately for solving all optimization problems [58]. The reason is because most algorithms are designed based on generalized concepts rather than domain knowledge specific to each problem. In this situation, hybridization technique for improving metaheuristic algorithms becomes important because it combines the desirable characteristics of different methods to mitigate their individual weaknesses. To design a good hybrid optimization algorithm, each algorithm’s deficiencies and privileges must be known to balance exploration and exploitation. In line with this, FA used as a heuristic is suitable for hybridizing with other general problem solvers because of its characteristics of multi-modality and fast convergence. For the SOS algorithm, the main advantage is that no specific parameters need to be set at the beginning of the algorithm. With this privilege, a hybrid algorithm with a few parameters can be obtained. Therefore, we design a novel hybrid MFA-SOS algorithm by combining the FA with SOS to solve the proposed multi-objective portfolio model.

Although many scholars have done researches on portfolio selection problem based on uncertainty theory, there are relatively less studies on uncertain multiobjective multiperiod portfolio selections, especially the semi-entropy is employed as a downside risk measure. On the other hand, inadequate exploration may cause FA to fall into a local optimum, although the FA has been shown to be effective. As a global optimizer, SOS can explore as much of the entire search space as possible, but the lack of directionality will cause the algorithm to waste more time searching for the optimal solution, thereby affecting the convergence speed. With the above issues, the contributions of this paper can be summarized as follows:

- We propose a new tri-objective multi-period mean-semientropy-skewness portfolio selection model, in which the uncertain semi-entropy is used as a risk measure. Besides, several real-world constraints are also taken into account, such as the transaction costs, cardinality, liquidity, budget, and bound constraint.
- We design a new MFA-SOS algorithm based on the hybridization of the firefly algorithm (FA) and symbiotic organism search algorithm (SOS) to solve the proposed model. To better balance exploration and exploitation, an improved search strategy and a poor-solution compensation strategy are proposed.

The remainder of this paper is organized as follows. Section 2 reviews some preliminaries about uncertainty theory. Section 3 gives the definition of uncertain semi-entropy and formulates an uncertain tri-objective multi-period mean-semientropy-skewness portfolio selection model. Section 4 develops the hybrid MFA-SOS algorithm for solving the proposed model. A numerical example is given in Section 5. Finally, Section 6 presents conclusions and future work.

2 Preliminaries

In this section, we recall some basic knowledge about uncertainty theory, which will be used throughout the article.

Definition 1 [16] Assume that Γ is a nonempty set and $\mathcal{L}$ is a σ -algebra on Γ . If a set function $\mathcal{M}$ is called an uncertain measure, then the following axioms hold:

Axiom 1 (Normality) $\mathcal{M}\{\Gamma\}=1$;

Axiom 2 (Duality) $\mathcal{M}\{\Lambda\} + \mathcal{M}\{\Lambda^c\}=1$ for any event $\Lambda \in \mathcal{L}$;

Axiom 3 (Subadditivity) For any given countable sequence $\Lambda_1, \Lambda_2, \dots$, we have

$$\mathcal{M}\left\{\bigcup_{i=1}^{\infty} \Lambda_i\right\} \leq \sum_{i=1}^{\infty} \mathcal{M}\{\Lambda_i\}.$$

Axiom 4 (Product) Assume that $(\Gamma_k, \mathcal{L}_k, \mathcal{M}_k)$ are uncertainty spaces for all $k = 1, 2, \dots, \infty$. Then, the product uncertain measure $\mathcal{M}$ satisfies the following relation

$$\mathcal{M}\left\{\prod_{k=1}^{\infty} \Lambda_k\right\} = \bigwedge_{k=1}^{\infty} \mathcal{M}_k\{\Lambda_k\}.$$

Here, Λ_k is randomly chosen events from $\mathcal{L}_k$ for $k = 1, 2, \dots$

2.1 Expected value

The expected value is the average value of the uncertain variable in the sense of the uncertainty measure, representing the size of the uncertain variable. The most general definition of the expected value of the uncertain variable is given below:

Definition 2 [16] Let ξ be an uncertain variable, then the expected value of ξ is defined by

$$E[\xi] = \int_0^{+\infty} \mathcal{M}\{\xi \geq r\} dr - \int_{-\infty}^0 \mathcal{M}\{\xi \leq r\} dr, \quad (1)$$

provided that at least one of the two integrals is finite.

Since the uncertain variables operate mainly in the form of inverse uncertainty distributions, Liu [18] proposed a formula to calculate the expected value of an uncertain variable via inverse uncertainty distributions.

Theorem 1 [18] *Let ξ be an uncertain variable with regular uncertainty distribution Φ . If the expected value exists, then*

$$E[\xi] = \int_0^1 \Phi^{-1}(\alpha) d\alpha. \quad (2)$$

Theorem 2 [18] *Let ξ be an uncertain variable. Then $\forall a, b \in \mathfrak{R}$, we have*

$$E[a\xi + b] = aE[\xi] + b. \quad (3)$$

2.2 Entropy

Definition 3 [18] Suppose that ξ is an uncertain variable with uncertainty distribution Φ . Then its entropy is defined by

$$H(\xi) = \int_{-\infty}^{+\infty} S(\Phi(x)) dx, \quad (4)$$

where $S(t) = -t \ln t - (1-t) \ln(1-t)$.

According to the Definition 3, entropy describes uncertainty that is greater than or less than the expected value. Entropy is one of the most common risk measures. However, it has the disadvantage that it cannot measure the downside risk of portfolio. Therefore, in the next section, we define the concept of uncertain semi-entropy as the downside risk measure of the portfolio.

2.3 Skewness

In portfolio selection, skewness measures the third order of return. The skewness becomes particularly important when the return distribution of the portfolio is asymmetric. The definition and characteristics of the uncertainty skewness are given below.

Definition 4 [20] Let ξ be an uncertain variable with finite expected value e . Then the skewness of ξ is defined by

$$S[\xi] = E[(\xi - e)^3]. \quad (5)$$

Theorem 3 [20] *Let ξ be an uncertain variable. Then $\forall a, b \in \mathfrak{R}$, we have*

$$S[a\xi + b] = a^3 S[\xi]. \quad (6)$$

Following this definition and property, the crisp form of a zigzag uncertain variable is given by Chen et al. [59].

Theorem 4 [59] *Let ξ be a zigzag uncertain variable denoted by $\mathcal{Z}(a, b, c)$ with the finite expected value e . Then, we have*

$$S[\xi] = \frac{(b-e)^4}{8(b-a)} - \frac{(a-e)^4}{8(b-a)} + \frac{(c-e)^4}{8(c-b)} - \frac{(b-e)^4}{8(c-b)}. \quad (7)$$

3 Problem formulation

In this section, we first give the definition and some characteristics of the semi-entropy of uncertain variables. Then, a new uncertain multi-objective multi-period portfolio optimization model is established by employing semi-entropy to measure the risk.

3.1 Semi-entropy of uncertain variables

Compared with common risk measures such as variance and entropy, the main advantage of semi-entropy is its ability to capture the downside risk that is more important for profit maximization. To describe only the uncertainty below the expected value, we give the definition and some characteristics of the uncertain semi-entropy. Referring to Definition 3, the uncertain semi-entropy is defined as below.

Definition 5 Suppose that ξ is a continuous uncertain variable with uncertainty distribution Φ and finite expected value e . Then its semi-entropy is defined as

$$H^-(\xi) = \int_{-\infty}^{+\infty} S(\Phi^-(x)) dx, \quad (8)$$

where $S(t) = -t \ln t - (1-t) \ln(1-t)$ and

$$\Phi^-(x) = \begin{cases} \Phi(x), & \text{if } x \leq e, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Since $S(0) = 0$, the semi-entropy can be rewritten as

$$H^-(\xi) = \int_{-\infty}^e S(\Phi(x)) dx, \quad (10)$$

Theorem 5 *Let ξ be a continuous uncertain variable with finite expected value e . Then for any real numbers a, b with $a \geq 0$, we have*

$$H^-(a\xi + b) = aH^-(\xi). \quad (11)$$

Proof If $a = 0$, then (11) is trivial. If $a > 0$, we have $E[a\xi + b] = ae + b$ and $\mathcal{M}\{a\xi + b \leq x\} = \Phi((x-b)/a)$ for any $x \in \mathfrak{R}$. Then it follows from the definition of semi-entropy that

$$\begin{aligned}
H^-(a\xi + b) &= \int_{-\infty}^{ae+b} S\{\Phi((x-b)/a)\}dx \\
&= \int_{-\infty}^e aS\{\Phi(x)\}dx \\
&= aH^-(\xi).
\end{aligned}$$

The proof is complete. $\square$

Theorem 6 Let $\xi \sim \mathcal{Z}(a, b, c)$ be a zigzag uncertain variable with expected value $e = \frac{a+2b+c}{4}$. Define $\zeta(x) = x^2 \ln(x) - (1-x)^2 \ln(1-x)$. Then the semi-entropy is

$$H^-(\xi) = \begin{cases} (b-a)\rho - (b-a)\zeta(\rho), & \text{if } e \leq b, \\ \frac{c-a}{2} - (c-b)\tau + (c-b)\zeta(\tau), & \text{otherwise.} \end{cases} \quad (12)$$

where $\rho = \frac{2b+c-3a}{8(b-a)}$ and $\tau = \frac{3c-a-2b}{8(c-b)}$.

Proof For zigzag uncertain variable $\xi \sim \mathcal{Z}(a, b, c)$, we have

$$\Phi^-(x) = \begin{cases} \frac{x-a}{2(b-a)}, & a \leq x \leq e \text{ and } e \in (a, b), \\ \frac{x-a}{2(b-a)}, & a \leq x \leq b \text{ and } e \in (b, c), \\ \frac{x+c-2b}{2(c-b)}, & b \leq x \leq e \text{ and } e \in (b, c), \\ 0, & \text{otherwise.} \end{cases}$$

The computing of semi-entropy breaks down into two cases. If $e \leq b$, the semi-entropy is

$$\begin{aligned}
H^-(\xi) &= \int_a^e S\left(\frac{x-a}{2(b-a)}\right)dx \\
&= \frac{2b+c-3a}{8} - \frac{(2b+c-3a)^2}{64(b-a)} \ln \frac{2b+c-3a}{8(b-a)} \\
&\quad + \frac{(6b-5a-c)^2}{64(b-a)} \ln \frac{6b-5a-c}{8(b-a)}. \quad (13)
\end{aligned}$$

Otherwise, if $e \geq b$, the semi-entropy is

$$\begin{aligned}
H^-(\xi) &= \int_a^b S\left(\frac{x-a}{2(b-a)}\right)dx + \int_b^e S\left(\frac{x+c-2b}{2(c-b)}\right)dx \\
&= \frac{c-a}{2} - \frac{3c-a-2b}{8} + \frac{(3c-a-2b)^2}{64(c-b)} \ln \frac{3c-a-2b}{8(c-b)} \\
&\quad - \frac{(a-6b+5c)^2}{64(c-b)} \ln \frac{a-6b+5c}{8(c-b)}. \quad (14)
\end{aligned}$$

Denote $\rho_{t,i} = \frac{2b_{t,i}+c_{t,i}-3a_{t,i}}{8(b_{t,i}-a_{t,i})}$ and $\tau = \frac{3c-a-2b}{8(c-b)}$ for simplification. Define $\zeta(x) = x^2 \ln(x) - (1-x)^2 \ln(1-x)$. Then the semi-entropy becomes

$$H^-(\xi) = \begin{cases} (b-a)\rho - (b-a)\zeta(\rho), & \text{if } e \leq b, \\ \frac{c-a}{2} - (c-b)\tau + (c-b)\zeta(\tau), & \text{otherwise.} \end{cases} \quad (15)$$

The proof is complete. $\square$

3.2 The uncertain mean-semi-entropy-skewness portfolio selection model

In this paper, we construct a mean-semi-entropy-skewness portfolio selection model under an uncertain environment. In the proposed model, the semi-entropy is used as the risk measure, and the skewness of portfolio is also taken into account. Simultaneously, many criteria such as the transaction cost, liquidity, cardinality, bound on holdings and capital budget are taken into account, so that the proposed model is more propitious to the real financial market.

We assume that the initial wealth W_0 is allocated among n securities. At the beginning of each investment period, the investor can readjust the investment proportions of securities for a good obtainment. Then, he could obtain the terminal wealth at the end of period T . To clearly describe the proposed model, the notations are given as follows:

$r_{t,i}$,	the uncertain return rate of security i at period t ;
$x_{t,i}$,	the investment proportion of security i at period t ;
W_t ,	the available wealth at the end of period t ;
$d_{t,i}$,	the transaction cost of security i at period t ;
$v_{t,i}$,	the lower bound of security i at period t ;
$\mu_{t,i}$,	the upper bound of security i at period t ;
m_t ,	the number of assets held at period t in the portfolio;
$z_{t,i}$,	a binary variable that takes a value of 1 if the i th security is included at period t and 0 otherwise.

3.2.1 Decision objectives

• Maximizing the terminal wealth.

For the transaction costs, we assume that the transaction costs at period t is a V-shape function of the difference between the t th period portfolio $x_t = (x_{t,1}, x_{t,2}, \dots, x_{t,n})$ and the $t-1$ th period portfolio $x_{t-1} = (x_{t-1,1}, x_{t-1,2}, \dots, x_{t-1,n})$. Thus, the transaction costs of security i at period t can be expressed by

$$D_{t,i} = d_{t,i}|x_{t,i} - x_{t-1,i}|. \quad (16)$$

Then, the net return rate of the portfolio at period t can be represented by

$$R_{t,N} = \sum_{i=1}^n (r_{t,i}x_{t,i} - d_{t,i}|x_{t,i} - x_{t-1,i}|). \quad (17)$$

Similar to [19], we suppose that the security returns $r_{t,i}$ ($i = 1, 2, \dots, n$; $t = 1, 2, \dots, T$) are zigzag uncertain variables, denoted by $\mathcal{Z}(a_{t,i}, b_{t,i}, c_{t,i})$, where $a_{t,i}$, $b_{t,i}$ and $c_{t,i}$ are real numbers and $a_{t,i} < b_{t,i} < c_{t,i}$.

The zigzag uncertainty distribution of $r_{t,i}$ is

$$\Phi(r_{t,i}) = \begin{cases} 0, & r_{t,i} \leq a_{t,i}, \\ \frac{r_{t,i}-a_{t,i}}{2(b_{t,i}-a_{t,i})}, & a_{t,i} \leq r_{t,i} \leq b_{t,i}, \\ \frac{r_{t,i}+c_{t,i}-2b_{t,i}}{2(c_{t,i}-b_{t,i})}, & b_{t,i} \leq r_{t,i} \leq c_{t,i}, \\ 1, & r_{t,i} \geq c_{t,i}. \end{cases} \quad (18)$$

Then the inverse uncertainty distributions of $r_{t,i}$ ($i = 1, 2, \dots, n$; $t = 1, 2, \dots, T$) are

$$\Phi^{-1}(\alpha_{t,i}) = \begin{cases} (1-2\alpha_{t,i})a_{t,i} + 2\alpha_{t,i}b_{t,i}, & 0 < \alpha_{t,i} < 0.5, \\ (2-2\alpha_{t,i})b_{t,i} + (2\alpha_{t,i}-1)c_{t,i}, & 0.5 \leq \alpha_{t,i} < 1. \end{cases} \quad (19)$$

Thus, from (2) and (19), we have

$$E[r_{t,i}] = \frac{a_{t,i} + 2b_{t,i} + c_{t,i}}{4}. \quad (20)$$

Since $r_{t,i}$ are zigzag uncertain variables for all $i = 1, 2, \dots, n$ and $t = 1, 2, \dots, T$, by the operational rules of uncertain variables, the return of the portfolio at period t is also a zigzag uncertain variable with the following form:

$$\sum_{i=1}^n x_{t,i} r_{t,i} = \mathcal{Z} \left(\sum_{i=1}^n x_{t,i} a_{t,i}, \sum_{i=1}^n x_{t,i} b_{t,i}, \sum_{i=1}^n x_{t,i} c_{t,i} \right). \quad (21)$$

Thus, we can easily obtain the expected value of the terminal wealth at the end of period T , which can be expressed as

$$E[W_T] = W_0 \prod_{t=1}^T \left[1 + \sum_{i=1}^n \left(\frac{a_{t,i} + 2b_{t,i} + c_{t,i}}{4} \right) x_{t,i} - \sum_{i=1}^n d_{t,i} |x_{t,i} - x_{t-1,i}| \right]. \quad (22)$$

• Minimizing the semi-entropy.

In the actual investment process, some investors regard low returns as undesirable, while high returns are accepted. In other words, investors only focus on the downside risk of portfolio, especially when the returns are asymmetrical. Thus, we use the uncertain semi-entropy to quantify the portfolio downside risk in this paper. According to Theorems 5 and 6, we obtain the cumulative risk over period T as follows:

$$H^-(r_{t,i}) = \begin{cases} \sum_{i=1}^T \left[\left(\sum_{i=1}^n b_{t,i} x_{t,i} - \sum_{i=1}^n a_{t,i} x_{t,i} \right) \rho_{t,i} - \left(\sum_{i=1}^n b_{t,i} x_{t,i} - \sum_{i=1}^n a_{t,i} x_{t,i} \right) \zeta(\rho_{t,i}) \right], \\ \quad a_{t,i} + c_{t,i} \leq 2b_{t,i}, \\ \sum_{i=1}^T \left[\frac{1}{2} \left(\sum_{i=1}^n c_{t,i} x_{t,i} - \sum_{i=1}^n a_{t,i} x_{t,i} \right) - \left(\sum_{i=1}^n c_{t,i} x_{t,i} - \sum_{i=1}^n b_{t,i} x_{t,i} \right) \tau_{t,i} \right. \\ \quad \left. + \left(\sum_{i=1}^n c_{t,i} x_{t,i} - \sum_{i=1}^n b_{t,i} x_{t,i} \right) \zeta(\tau_{t,i}) \right], \\ \quad a_{t,i} + c_{t,i} \geq 2b_{t,i}. \end{cases} \quad (23)$$

$$\text{where } \rho_{t,i} = \frac{\sum_{i=1}^n (2b_{t,i} + c_{t,i} - 3a_{t,i})}{8 \sum_{i=1}^n (b_{t,i} - a_{t,i})}, \quad \tau_{t,i} = \frac{\sum_{i=1}^n (3c_{t,i} - a_{t,i} - 2b_{t,i})}{8 \sum_{i=1}^n (c_{t,i} - b_{t,i})} \text{ and } \zeta(x) = x^2 \ln(x) - (1-x)^2 \ln(1-x).$$

• Maximizing the skewness.

Markowitz et al. [60] showed that asset returns that are not often symmetrically distributed typically have heavy tailed. Several researchers have investigated the portfolio selection problem by incorporating the third moments of return distributions into the analyses [61, 62]. Therefore, in this paper, we consider the skewness of the returns in portfolio selection. According to Theorems 3 and 4, the cumulative skewness over T periods can be expressed as below:

$$S(r_{t,i}) = \frac{1}{2048} \sum_{t=1}^T \left[\frac{[\sum_{i=1}^n (2b_{t,i} - a_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (b_{t,i} - a_{t,i}) x_{t,i}} - \frac{[\sum_{i=1}^n (3a_{t,i} - 2b_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (b_{t,i} - a_{t,i}) x_{t,i}} + \frac{[\sum_{i=1}^n (3c_{t,i} - a_{t,i} - 2b_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (c_{t,i} - b_{t,i}) x_{t,i}} - \frac{[\sum_{i=1}^n (2b_{t,i} - a_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (c_{t,i} - b_{t,i}) x_{t,i}} \right] \quad (24)$$

3.2.2 Investment constraints

• Liquidity constraint.

The liquidity risk of portfolio measures the possibility of converting an investment into cash without any significant loss, which is one of the most concerning issues for investors in actual investment. Here, the illiquidity index of security proposed by Amihud [63] is used to measure the liquidity risk. We suppose that the illiquidity index $l_{t,i}$ ($i = 1, 2, \dots, n$; $t = 1, 2, \dots, T$) are zigzag uncertain variables, denoted by $\mathcal{Z}(a_{t,i}^l, b_{t,i}^l, c_{t,i}^l)$. Then, the expected liquidity risk of portfolio $E(L_t)$ at period t can be given by

$$E(L_t) = \sum_{i=1}^n x_{t,i} \frac{a_{t,i}^l + 2b_{t,i}^l + c_{t,i}^l}{4}. \quad (25)$$

• Cardinality constraint.

In the real world, investors may wish to restrict the number of securities in a portfolio at each period to control the transaction costs and reduce the complexity of the portfolio. Thus, in this paper, we assume that the investor intends to hold m_t securities in the portfolio at period t . Then, the cardinality constraints about the portfolio at period t can be expressed by

$$\sum_{i=1}^n z_{t,i} = m_t, \quad t = 1, 2, \dots, T, \quad (26)$$

where $z_{t,i} \in \{0, 1\}$ is a binary variable that controls whether security i at period t should be selected in the portfolio or not.

- **Bound constraint.**

In the real investment, investors may specify the lower and the upper investment proportion for each asset. Then, at period t ($t = 1, \dots, T$), the proportion constraint is given by

$$v_{t,i} z_{t,i} \leq x_{t,i} \leq \mu_i z_{t,i}. \quad (27)$$

- **Capital budget constraint.**

Ideally, all available capital should be completely invested so that the capital budget constraint can be expressed as

$$\sum_{i=1}^n x_{t,i} = 1, \quad t = 1, 2, \dots, T. \quad (28)$$

3.2.3 Model formulation

Based on the above analysis, we suppose that the investor's investment goals are maximizing terminal wealth, minimizing risk and maximizing skewness. Meanwhile, the investor takes some realistic constraints into consideration, such as liquidity constraint, cardinality constraint and so on. Then, the multiobjective multiperiod mean-semi-entropy-skewness portfolio model is formulated as follows:

$$\text{If } a_{t,i} + c_{t,i} \leq 2b_{t,i},$$

$$\begin{cases} \max & W_0 \prod_{t=1}^T \left[1 + \sum_{i=1}^n \left(\frac{a_{t,i} + 2b_{t,i} + c_{t,i}}{4} \right) x_{t,i} - \sum_{i=1}^n d_{t,i} |x_{t,i} - x_{t-1,i}| \right] \\ \min & \sum_{t=1}^T \left[\left(\sum_{i=1}^n b_{t,i} x_{t,i} - \sum_{i=1}^n a_{t,i} x_{t,i} \right) \rho_{t,i} - \left(\sum_{i=1}^n b_{t,i} x_{t,i} - \sum_{i=1}^n a_{t,i} x_{t,i} \right) \zeta(\rho_{t,i}) \right] \\ \max & \frac{1}{2048} \sum_{t=1}^T \left[\frac{[\sum_{i=1}^n (2b_{t,i} - a_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (b_{t,i} - a_{t,i}) x_{t,i}} - \frac{[\sum_{i=1}^n (3a_{t,i} - 2b_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (b_{t,i} - a_{t,i}) x_{t,i}} \right. \\ & \quad \left. + \frac{[\sum_{i=1}^n (3c_{t,i} - a_{t,i} - 2b_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (c_{t,i} - b_{t,i}) x_{t,i}} - \frac{[\sum_{i=1}^n (2b_{t,i} - a_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (c_{t,i} - b_{t,i}) x_{t,i}} \right] \\ \text{s.t.} & \text{Constraints for (25) - (28),} \\ & i = 1, 2, \dots, n, \quad t = 1, 2, \dots, T. \end{cases} \quad (29)$$

$$\text{where } \rho_{t,i} = \frac{\sum_{i=1}^n (2b_{t,i} + c_{t,i} - 3a_{t,i})}{8 \sum_{i=1}^n (b_{t,i} - a_{t,i})} \text{ and } \zeta(x) = x^2 \ln(x) - (1-x)^2 \ln(1-x).$$

$$\text{If } a_{t,i} + c_{t,i} \geq 2b_{t,i},$$

$$\begin{cases} \max & W_0 \prod_{t=1}^T \left[1 + \sum_{i=1}^n \left(\frac{a_{t,i} + 2b_{t,i} + c_{t,i}}{4} \right) x_{t,i} - \sum_{i=1}^n d_{t,i} |x_{t,i} - x_{t-1,i}| \right] \\ \min & \sum_{t=1}^T \left[\frac{1}{2} \left(\sum_{i=1}^n c_{t,i} x_{t,i} - \sum_{i=1}^n a_{t,i} x_{t,i} \right) - \left(\sum_{i=1}^n c_{t,i} x_{t,i} - \sum_{i=1}^n b_{t,i} x_{t,i} \right) \tau_{t,i} \right. \\ & \quad \left. + \left(\sum_{i=1}^n c_{t,i} x_{t,i} - \sum_{i=1}^n b_{t,i} x_{t,i} \right) \zeta(\tau_{t,i}) \right] \\ \max & \frac{1}{2048} \sum_{t=1}^T \left[\frac{[\sum_{i=1}^n (2b_{t,i} - a_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (b_{t,i} - a_{t,i}) x_{t,i}} - \frac{[\sum_{i=1}^n (3a_{t,i} - 2b_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (b_{t,i} - a_{t,i}) x_{t,i}} \right. \\ & \quad \left. + \frac{[\sum_{i=1}^n (3c_{t,i} - a_{t,i} - 2b_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (c_{t,i} - b_{t,i}) x_{t,i}} - \frac{[\sum_{i=1}^n (2b_{t,i} - a_{t,i} - c_{t,i}) x_{t,i}]^4}{\sum_{i=1}^n (c_{t,i} - b_{t,i}) x_{t,i}} \right] \\ \text{s.t.} & \text{Constraints for (25) - (28),} \\ & i = 1, 2, \dots, n, \quad t = 1, 2, \dots, T. \end{cases} \quad (30)$$

$$\text{where } \tau_{t,i} = \frac{\sum_{i=1}^n (3c_{t,i} - a_{t,i} - 2b_{t,i})}{8 \sum_{i=1}^n (c_{t,i} - b_{t,i})} \text{ and } \zeta(x) = x^2 \ln(x) - (1-x)^2 \ln(1-x).$$

4 The proposed algorithm

4.1 The basic FA

FA is a nature-inspired algorithm proposed by Yang [41] in 2008, which comes from the simplification and simulation of firefly group behavior. FA follows three rules: (I) Fireflies are unisex, (II) the attraction of a firefly is proportional to its brightness, (III) the brightness of a firefly is determined by the objective function.

The movement equation of firefly i can be expressed as

$$X_i^{t+1} = X_i^t + \beta_{i,j}(r_{ij})(X_j^t - X_i^t) + \alpha_t \epsilon_i^t, \quad (31)$$

$$\beta_{i,j}(r_{ij}) = \beta_0 e^{-\gamma r_{ij}^2}, \quad (32)$$

$$r_{ij} = \|X_i - X_j\| = \sqrt{\sum_{k=1}^D (X_{i,k} - X_{j,k})^2}. \quad (33)$$

Here, $\beta_{i,j}(r_{ij})$ represents the attractiveness between firefly i and firefly j , which is determined by the distance between two fireflies, i.e., r_{ij} . β_0 is the maximum attraction, which is the attraction at the light source ($r = 0$). γ is the light absorption coefficient and represents the change of attraction. D is the dimension of the objective function to be optimized. $X_{i,k}$ and $X_{j,k}$ are the k th component elements of X_i and X_j , respectively. α_t is a random number, and ϵ_i^t is a random number vector from a gaussian, a uniform distribution, or some other distributions.

4.2 The basic SOS

SOS is an evolutionary search method developed by Cheng and Prayogo in 2014 [51]. The SOS algorithm simulates the symbiotic interaction strategies of organisms in ecosystems. Similar to other algorithms, SOS has operators that update the positions of organisms in each iteration. These operators, which resemble a biological interaction model, are introduced as Mutualism, Commensalism and Parasitism. The mathematical implementation of SOS can be explained as follows.

- **Mutualism phase.**

$$X_i^{t+1} = X_i^t + \text{rand}(0, 1) * (X_{best}^t - MV * BF_1), \quad (34)$$

$$X_j^{t+1} = X_j^t + \text{rand}(0, 1) * (X_{best}^t - MV * BF_2), \quad (35)$$

where

$$MV = \frac{X_i^t + X_j^t}{2},$$

$$BF_1/BF_2 = \text{round}[\text{rand}(0, 1)] + 1.$$

Here, X_j is a organism randomly selected from the current population to interact with X_i . $\text{rand}(0, 1)$ in (34) and (35) is a vector of random numbers. And the benefit factors (BF_1 and BF_2) are determined randomly as either 1 or 2.

- **Commensalism phase.**

$$X_i^{t+1} = X_i^t + \text{rand}(-1, 1) * (X_{best}^t - X_j^t), \quad (36)$$

where X_j is a organism selected randomly from the current population.

- **Parasitism phase.** In this phase, a “Parasite-Vector” is created by duplicating organism X_i , then modifying the randomly selected dimensions by a random number. Organism X_j is randomly selected from the ecosystem and serves as a victim for the “Parasite-Vector”. Then, both “Parasite-Vector” and X_j are evaluated to measure their fitness. If the fitness value of “Parasite-Vector” is better than X_j , “Parasite-Vector” will kill organism X_j and become an individual in the ecosystem. Otherwise, organism X_j will have immunity from the parasite and “Parasite-Vector” will no longer live.

4.3 The hybrid MFA-SOS algorithm

In this paper, we combine the desirable characteristics of FA and SOS algorithms to design a new hybrid multi-objective algorithm called MFA-SOS. FA is suitable for hybridization with other optimizers due to its multi-modality and fast convergence characteristics. However, when encountering extreme values, the exploration capability of FA may have limitations. Fireflies may fall into a local optimum because they cannot cross the extreme point. The main advantage

of SOS is that it does not need any specific parameters at the beginning of the algorithm to be set. With this privilege, the hybrid algorithm can be obtained with a few parameters. Although SOS can explore as much of the entire search space as possible, the convergence speed is undesirable due to the lack of directionality. Combining the privileges of FA and SOS and avoiding their deficiencies, an improved search strategy and a poor-solution compensation strategy are proposed in the hybrid MFA-SOS algorithm to improve the quality of the solutions.

4.3.1 Population initialization

Population initialization is one of the most important tasks of metaheuristic algorithms because the initial population will affect the convergence speed and the quality of the final solutions. First, a population comprised of SN fireflies is generated by random initialization, where each firefly has $n * T$ dimensions representing n securities and T periods. The population is expressed as

$$\vec{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_{SN} \end{bmatrix} = \begin{bmatrix} x_{1,1}^1 & x_{1,2}^1 & \cdots & x_{1,n}^1 & \cdots & x_{T,1}^1 & x_{T,2}^1 & \cdots & x_{T,n}^1 \\ x_{1,1}^2 & x_{1,2}^2 & \cdots & x_{1,n}^2 & \cdots & x_{T,1}^2 & x_{T,2}^2 & \cdots & x_{T,n}^2 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ x_{1,1}^{SN} & x_{1,2}^{SN} & \cdots & x_{1,n}^{SN} & \cdots & x_{T,1}^{SN} & x_{T,2}^{SN} & \cdots & x_{T,n}^{SN} \end{bmatrix}.$$

To construct the cardinality constraint, we introduce a binary matrix Z ,

$$\vec{Z} = \begin{bmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_{SN} \end{bmatrix} = \begin{bmatrix} z_{1,1}^1 & z_{1,2}^1 & \cdots & z_{1,n}^1 & \cdots & z_{T,1}^1 & z_{T,2}^1 & \cdots & z_{T,n}^1 \\ z_{1,1}^2 & z_{1,2}^2 & \cdots & z_{1,n}^2 & \cdots & z_{T,1}^2 & z_{T,2}^2 & \cdots & z_{T,n}^2 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ z_{1,1}^{SN} & z_{1,2}^{SN} & \cdots & z_{1,n}^{SN} & \cdots & z_{T,1}^{SN} & z_{T,2}^{SN} & \cdots & z_{T,n}^{SN} \end{bmatrix},$$

where all the elements in the matrix Z are binary variables with values of 0 or 1. Finally, the initial population P is obtained by multiplying the corresponding elements in the matrices X and Z .

4.3.2 The improved search strategy

Based on the FA framework, two components are included in the proposed search strategy. One is the dominated movement of individuals, inheriting from the search strategy of the FA algorithm. The other is the movement of the non-dominated individual, which is borrowed from the SOS algorithm. FA depends primarily on the variation of light intensity and the formulation of attractiveness. In FA, a dimmer firefly is attracted by a brighter one, so as to find the optimal solution quickly. However, some studies [64, 65] suggest that incomplete exploration may lead to premature convergence, while encountering local extrema. Thus, FA is

used as a local search operator in the hybrid algorithm to avoid this deficiency. In the proposed MFA-SOS, the fitness of firefly is determined by the objective functions. The direction and brightness of fireflies depend on the Pareto dominance relationship between them. In each iteration of the MFA-SOS algorithm, each firefly X_i compares its brightness with the other fireflies. In each comparison, if firefly X_i is dominated by firefly X_j , firefly X_i will move towards firefly X_j according to FA movement strategy (described in (31)–(33)). Since non-dominated fireflies may not be updated, search strategies of SOS and a chaotic map are introduced for non-dominated fireflies.

According to the search strategy of SOS, the activity among non-dominated solutions in the mutualism phase is described by (34)–(35). Here, the benefit factors (BF_1 and BF_2) represent the level of benefit to each solution. MV represents the relationship characteristic between X_i and X_j . Part of the equation, $(X_{best}^t - MV * BF_1)$, reflects the mutualistic effort to increase their survival advantage. In the firefly population, the brighter firefly, namely the closer to the optimal solution, is considered to have a greater survival advantage. However, the direction and step length of each individual moving towards the optimal individual is relatively random in (34)–(35), and the brightness characteristics of the fireflies are not reflected during the update process. This phenomenon may cause excellent fireflies to waste more time converging to the optimal position. To overcome this shortcoming, parameter β inherited from FA is introduced into the SOS learning formulation. From the algorithm perspective, the introduction of parameter β has a more versatile adjustment for the step length and direction of the firefly movement, which usually leads to higher mobility and thus the search space is explored more efficiently. Then, (34) and (35) are replaced with (37) and (38), respectively.

$$X_i^{t+1} = X_i^t + rand(0, 1) * \beta_1 * (X_{best}^t - MV * BF_1), \quad (37)$$

$$X_j^{t+1} = X_j^t + rand(0, 1) * \beta_2 * (X_{best}^t - MV * BF_2), \quad (38)$$

where β_1 is the attractiveness between X_{best} and X_i , and β_2 is the attractiveness between X_{best} and X_j . Both β_1 and β_2 are calculated from (32)–(33). Subsequently, individual X_j is randomly selected from the current population and commensal symbiosis with X_i . X_i attempts to benefit from the interaction but X_j is not affected. In this commensalism phase, X_i is updated by (36), which is inherited from the second phase of the basic SOS. Due to the existence of several random variables in the learning formula, some fireflies may be far away from the optimal solution. Therefore, a chaotic map mechanism is introduced to improve the global search mobility for robust global optimization. In this paper, a sine map [66] is employed due

to its simplicity. The sine map is described as

$$X_i^{t+1} = \frac{a}{4} \sin(\pi X_i^t). \quad (39)$$

Finally, the location of firefly i is updated only if the new solution dominates the previous one.

4.3.3 Poor-solution compensation strategy

Exploration and exploitation are two important factors for any optimization algorithm, and they are considered the basis for successfully solving optimization problems. In the search process of an algorithm, a more global search should be conducted first. This can allow the algorithm to have a strong search ability in the early stages. After sufficient exploration, the process should move on to local search. In this paper, a poor-solution compensation strategy is introduced. After the population is updated, all fireflies are sorted from good to bad based on the fast non-dominated sorting algorithm [67] and crowding distance. Afterwards a parameter q is introduced in this paper to determine the number of fireflies that need to be compensated for. In each iteration, the last q fireflies in the population are treated as poor fireflies, and they are compensated for in a mutated way. Then, parameter q can be calculated as

$$q = \lfloor rand(0, 1) * SN * \frac{1}{t} \rfloor + 1. \quad (40)$$

where SN is the population size, t is the current number of iterations, and $\lfloor \cdot \rfloor$ represents the rounding operation. It can be seen from (40) that q decreases in the iterative process. In other words, at the beginning, the algorithm tends to explore the search space better, with a large q . After enough exploration has been performed, the process moves on to the local search that is exploitation, with a small q .

Finally, the stepwise implementation of the proposed MFA-SOS is narrated below:

Step 1: Initialize the population.

Step 2: Evaluate the fitness of each firefly. According to (22)–(24), the values of the objective functions associated with each firefly X_i , $\forall i = [1, SN]$ are calculated.

Step 3: Perform the fast non-dominated sorting algorithm. Individuals in the population are sorted to find the optimal individual X_{best} by the fast non-dominated sorting algorithm. Then, the individuals whose rank = 1 are the non-dominated solutions, and the other individuals are the dominated solutions.

Step 4: Update the population. According to the improved search strategy, the population of fireflies is divided into two sub-populations, namely dominated fireflies and non-dominated fireflies.

For the dominated firefly individuals, the movement follows the flashing patterns of fireflies described as (31)–(33). For the non-dominated fireflies, the update formulations are expressed as (36)–(39).

Step 5: Implement the poor-solution compensation strategy.

Step 6: Select fireflies for the next generation. An intermediate population is generated by merging the current population with the previous population, which is denoted by $P_t^{temp} = [P_t, P_t^{curr}]^T$. Then, SN fireflies are selected for the next generation based on the fast non-dominated sorting algorithm and crowding distance.

Repeat Steps 4–6 until the maximum number of iterations is reached.

4.4 Constraint handling

The penalty function method [68, 69] is one of the common methods for handling constraints. Thus, in this paper, the penalty function method is employed to deal with the constraint (25), i.e., liquidity constraint.

In addition, to deal with other constraints in the proposed model, the following repair mechanism is performed. First, the vectors $x = \{x_1, x_2, \dots, x_T\}$ with $x_t = (x_{t,1}, x_{t,2}, \dots, x_{t,n})$, and $z = \{z_1, z_2, \dots, z_T\}$ with $z_t = (z_{t,1}, z_{t,2}, \dots, z_{t,n})$ are introduced to represent the portfolio for all $t = 1, 2, \dots, T$. Here, z_t is a binary vector specifying whether a particular security participates in the portfolio at period t , where $z_{t,i} \in \{0, 1\}$, $i = 1, 2, \dots, n$; $t = 1, 2, \dots, T$. If the number of securities in the portfolio in period t exceeds the maximum allowable value m_t , the $(n - m_t)$ securities with the smallest weight in x_t are deleted. Otherwise, if the number of securities in the portfolio in period t is less than m_t , a security is randomly selected from the securities that is not in portfolio and a random weight is given. This step is repeated until the cardinality constraint, i.e., constraint (26), is satisfied.

Finally, in order to satisfy the capital budget constraint, the following normalization operation is performed

$$x_{t,i}^{adjust} = \frac{x_{t,i} z_{t,i}}{\sum_{i=1}^n x_{t,i} z_{t,i}}, \quad i = 1, 2, \dots, n; t = 1, 2, \dots, T. \quad (41)$$

Since the adjusted investment proportion may not satisfy the bound constraints, we discuss the following three scenarios.

Scenario 1: If the investment proportion is less than the lower bound and there is no restriction on the

upper bound, then the investment proportion is adjusted to

$$x_{t,i}^{adjust} = v_{t,i} z_{t,i} + \frac{x_{t,i} z_{t,i}}{\sum_{i=1}^n x_{t,i} z_{t,i}} \left(1 - \sum_{i=1}^n v_{t,i} z_{t,i} \right), \quad i = 1, 2, \dots, n; t = 1, 2, \dots, T. \quad (42)$$

Scenario 2: If the investment proportion is greater than the upper bound and there is no restriction on the lower bound, then the investment proportion is adjusted to

$$x_{t,i}^{adjust} = \mu_{t,i} z_{t,i} + \frac{x_{t,i} z_{t,i}}{\sum_{i=1}^n x_{t,i} z_{t,i}} \left(\sum_{i=1}^n \mu_{t,i} z_{t,i} - 1 \right), \quad i = 1, 2, \dots, n; t = 1, 2, \dots, T. \quad (43)$$

Scenario 3: If the lower and upper bounds are both given, then the investment proportion is adjusted to

$$x_{t,i}^{adjust} = v_{t,i} z_{t,i} + \frac{x_{t,i} z_{t,i}}{\sum_{i=1}^n x_{t,i} z_{t,i}} \left(\mu_{t,i} z_{t,i} - \sum_{i=1}^n v_{t,i} z_{t,i} \right), \quad i = 1, 2, \dots, n; t = 1, 2, \dots, T. \quad (44)$$

5 Numerical example

In this section, we present a numerical experiment introduced by Jin et al. [70] to illustrate the application of the proposed model and demonstrate the effectiveness of the designed algorithm.

5.1 Parameter setting

In this example, we assume that the investor's initial wealth $W_0 = 1$. The investor wants to construct a three-period ($T = 3$) investment strategy among 10 securities. Here, the return rates and illiquidity indexes characterized by zigzag uncertain variables, are shown in Tables 1 and 2. Moreover, the initial portfolio is $x_{0,i} = 0.1$ for 10 securities (i.e., $i = 1, 2, \dots, 10$). The transaction costs at the three periods are $d_{1,i} = 0.001$, $d_{2,i} = 0.002$, and $d_{3,i} = 0.003$, $i = 1, 2, \dots, 10$. It is also assumed that $\gamma_t = -0.008$ for the portfolio model. The lower and upper bounds are $v = 0.05$ and $\mu = 0.5$, respectively. The number of assets including the portfolio at each period t is $m_t = 5$, $t = 1, 2, 3$.

Additionally, to evaluate the performance of the designed MFA-SOS, we compare it with the basic MOFA [44], MOSOS [51], non-dominated sorting genetic algorithm II (NSGA-II) [67], multiobjective evolutionary algorithm based on decomposition (MOEA/D) [71], Pareto archived evolution strategy II (PESA-II) [72], multiobjective whale optimization algorithm (MOWOA) [73],

Table 1 The zigzag uncertain returns of 10 securities

T	Security i	AMKR	BLDR	CBCP	CCOI	CLNE	CMRX	COKE	DLTR	EFII	ENTG
$t=1$	$a_{1,i}$	-0.1407	-0.1357	0.0288	-0.0941	-0.3313	-0.6198	-0.0565	0.0734	-0.0765	-0.0787
	$b_{1,i}$	0.0160	0.0370	0.0893	0.0180	0.0003	0.0143	0.0169	0.0959	0.0102	0.0070
	$c_{1,i}$	0.1608	0.6482	0.5333	0.1942	0.1137	0.1639	0.8238	0.4696	0.3970	0.1439
$t=2$	$a_{2,i}$	-0.0217	-0.0468	-0.0345	0.0594	-0.3560	-0.1923	-0.3312	-0.0516	-0.0846	-0.0785
	$b_{2,i}$	-0.0087	0.0178	0.0142	0.2025	0.0153	0.0266	0.0231	0.0061	0.0231	0.0058
	$c_{2,i}$	0.1313	0.1242	0.4386	0.2487	0.1948	0.7936	0.2487	0.3229	0.5269	0.7914
$t=3$	$a_{3,i}$	-0.1395	-0.0867	-0.3042	-0.0932	-0.0596	-0.0839	0.0861	-0.0587	-0.0453	-0.0073
	$b_{3,i}$	0.0131	0.0100	0.0030	0.0085	0.0063	-0.0042	0.0959	0.0108	0.0168	0.1767
	$c_{3,i}$	0.1837	0.0959	0.5985	0.3220	0.2516	0.0026	0.2793	0.1104	0.0231	0.2222

multiobjective crow search algorithm (MOCSA) [74], multiobjective fruit fly optimization algorithm (MOFOA) [75]. The compared algorithms are briefly described as below:

- (1) MOFA is a nature-inspired algorithm that simulates the population behavior of fireflies. In MOFA, individuals with high brightness are attracted to individuals with low brightness, which makes them converge to the optimal position quickly. However, MOFA is readily exposed to local optimization, when a local extremum is encountered.
- (2) MOSOS is a robust and powerful metaheuristic algorithm, which simulates the symbiotic interaction adopted by organisms to survive and propagate in an ecosystem. A main advantage of the MOSOS algorithm is that algorithm operations require no specific algorithm parameters. However, the lack of directionality during the movement affects the convergence speed.
- (3) The NSGA-II, MOEA/D and PESA-II are all multi-objective evolutionary algorithms. The advantage of NSGA-II is that the solution set has good distribution, but the diversity of the solution set is not satisfactory in high-dimensional problems. For MOEA/D, a major characteristic is that a scalar objective local search can be used in each subproblem in a natural way since its task is to optimize a scalar objective subproblem. PESA-II has good convergence, especially in high-dimensional problems. However, only one individual can be selected at a time for the selection operator, which consumes considerable computational time.
- (4) The MOWOA, MOCSA and MOFOA are all swarm-based intelligence algorithms that imitate the behavior of whale, crow and fruit fly populations in nature, respectively. The striking features of the MOWOA are its successful execution even with a small number of parameters, but sometimes it may get trapped into local optima and fails to apply the global search exhaustively. A significant feature of MOCSA is that it contains a memory that good solutions are memorized, which means that the best positions found so far at each iteration will be used directly to find better positions. Similar to MOSOS and MOWOA, MOFOA also has a small number of adjustable parameters, but

Table 2 Uncertain illiquidity indexes at each period

T	Security i	AMKR	BLDR	CBCP	CCOI	CLNE	CMRX	COKE	DLTR	EFII	ENTG
$t=1$	$a_{1,i}^l$	0.0020	0.0022	0.0020	0.0020	0.0018	0.0061	0.0038	0.0079	0.0013	0.0009
	$b_{1,i}^l$	0.0030	0.0063	0.0030	0.0030	0.0028	0.0071	0.0057	0.0098	0.0019	0.0020
	$c_{1,i}^l$	0.0043	0.0095	0.0055	0.0035	0.0078	0.0089	0.0066	0.0142	0.0023	0.0050
$t=2$	$a_{2,i}^l$	0.0049	0.0031	0.0010	0.0010	0.0013	0.0014	0.0011	0.0061	0.0011	0.0009
	$b_{2,i}^l$	0.0055	0.0044	0.0037	0.0030	0.0019	0.0032	0.0020	0.0120	0.0018	0.0030
	$c_{2,i}^l$	0.0070	0.0094	0.0059	0.0037	0.0040	0.0072	0.0050	0.0165	0.0042	0.0040
$t=3$	$a_{3,i}^l$	0.0017	0.0010	0.0011	0.0007	0.0022	0.0039	0.0020	0.0059	0.0013	0.0016
	$b_{3,i}^l$	0.0052	0.0054	0.0026	0.0042	0.0032	0.0080	0.0042	0.0123	0.0019	0.0060
	$c_{3,i}^l$	0.0100	0.0086	0.0047	0.0054	0.0045	0.0181	0.0051	0.0156	0.0041	0.0075

Table 3 The main parameters' setting of algorithms

Algorithm	Parameters	Value
MFA-SOS	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	Attractiveness β_0	0.1
	Absorption parameter γ	0.0001
MOFA	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	Attractiveness β_0	0.1
	Absorption parameter γ	0.0001
MOSOS	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
NSGA-II	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	Crossover probability p_c	0.9
	Mutation probability p_m	0.1
MOEA/D	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	Crossover parameter	0.5
PESA-II	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	Crossover probability p_c	0.9
	Mutation probability p_m	0.1
	Number of grids per dimension	7
	Grid inflation factor	0.1
MOWOA	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	The shape of logarithmic spiral parameter b	1.2
MOCSA	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100
	Awareness probability AP	0.2
	Flight length fl	2
MOFOA	Population size	100
	Maximum number of generations	300
	Archive size $nArchive$	100

the optimization speed of MOFOA slows down with iteration leading to premature convergence.

In the following experiments, the parameter settings of nine algorithms are shown in Table 3. In our experiments, each algorithm was independently run approximately 20 times for each instance.

5.2 Experimental results

In this section, we present several cases with different cardinality constraints to illustrate the effectiveness of the proposed algorithm. In the cases of $m_t = 5$, $m_t = 7$ and $m_t = 9$, the mean (mean), standard deviation (std), maximum (max) and minimum (min) of the cumulative

Table 4 Performance comparisons in three periods under the cardinality constraint $m_t = 5$

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$t = 1$	Wealth	mean	1.178244	1.154262	1.176924	1.145279	1.148358	1.170906	1.150300	1.145070	1.144076
		std	0.001127	0.005206	0.001271	0.004037	0.012381	0.004046	0.020191	0.013298	0.005220
		max	1.179842	1.165872	1.179708	1.151958	1.166860	1.176305	1.175260	1.168447	1.153224
		min	1.175776	1.144817	1.174222	1.135455	1.113654	1.159701	1.104708	1.124216	1.130791
	Semi-entropy	mean	0.075524	0.064385	0.075942	0.070083	0.062041	0.068725	0.064586	0.068957	0.072330
		std	0.000682	0.003240	0.000868	0.002295	0.005104	0.006533	0.008824	0.005822	0.003218
		max	0.076338	0.069949	0.077357	0.073926	0.070791	0.079547	0.080482	0.077156	0.077541
		min	0.073879	0.059066	0.074441	0.066896	0.051973	0.053297	0.047911	0.057838	0.067423
	Skewness	mean	0.010110	0.004596	0.009886	0.004847	0.003579	0.006586	0.004833	0.004574	0.005231
		std	0.000299	0.000900	0.000289	0.000545	0.000970	0.001581	0.002291	0.001372	0.000788
		max	0.010840	0.006342	0.010477	0.005944	0.005097	0.009571	0.008345	0.007444	0.006652
		min	0.009558	0.003271	0.009462	0.003888	0.001992	0.002895	0.001249	0.002453	0.003916
$t = 2$	Wealth	mean	1.295447	1.298123	1.291983	1.270216	1.278607	1.306111	1.279211	1.270934	1.267800
		std	0.003018	0.009788	0.003043	0.008132	0.024309	0.009079	0.023912	0.018161	0.009272
		max	1.301683	1.314066	1.297237	1.282153	1.310030	1.319483	1.309009	1.301891	1.278526
		min	1.291242	1.276427	1.285086	1.242054	1.222428	1.280718	1.216649	1.242791	1.240237
	Semi-entropy	mean	0.133004	0.139779	0.132270	0.141830	0.129326	0.133694	0.130942	0.138884	0.141845
		std	0.001280	0.007094	0.001305	0.003588	0.005838	0.007273	0.012847	0.009530	0.005280
		max	0.135292	0.151559	0.134929	0.148578	0.140962	0.145324	0.154115	0.156980	0.152453
		min	0.130677	0.127468	0.129620	0.133478	0.119706	0.117496	0.113658	0.121127	0.131034
	Skewness	mean	0.015480	0.009791	0.014817	0.009560	0.006291	0.012027	0.008985	0.008927	0.009289
		std	0.000383	0.001784	0.000388	0.000515	0.001067	0.001580	0.003474	0.001548	0.000851
		max	0.016340	0.012947	0.015461	0.010680	0.008654	0.014599	0.015963	0.011837	0.010395
		min	0.014689	0.006332	0.013863	0.008115	0.004404	0.009953	0.003815	0.006417	0.007370
$t = 3$	Wealth	mean	1.424090	1.408155	1.411806	1.332489	1.377481	1.391519	1.355110	1.331146	1.331614
		std	0.005619	0.014938	0.008472	0.012635	0.034997	0.014395	0.055109	0.028913	0.013730
		max	1.432385	1.429565	1.421204	1.350832	1.427424	1.412693	1.424596	1.371665	1.349226
		min	1.411037	1.372869	1.392861	1.289711	1.310872	1.358951	1.245302	1.282724	1.296931
	Semi-entropy	mean	0.152696	0.170362	0.153309	0.182488	0.165463	0.160458	0.161486	0.176688	0.178367
		std	0.001374	0.006188	0.001323	0.006037	0.007769	0.006301	0.007999	0.013161	0.007923
		max	0.154918	0.181376	0.155735	0.191980	0.177557	0.169250	0.180091	0.203231	0.192618
		min	0.149644	0.161245	0.150748	0.168992	0.152229	0.150931	0.150610	0.160180	0.161839
	Skewness	mean	0.015656	0.009943	0.014974	0.009687	0.006433	0.011400	0.009070	0.009026	0.009397
		std	0.000384	0.001770	0.000391	0.000545	0.001058	0.001673	0.003522	0.001551	0.000852
		max	0.016526	0.013095	0.015627	0.010801	0.008691	0.014599	0.016048	0.011934	0.010545
		min	0.014872	0.006501	0.014000	0.008183	0.004564	0.008110	0.003819	0.006477	0.007479

wealth, risk and skewness for three periods are listed in Tables 4, 5 and 6. The best results are marked in bold. From Tables 4–6, we can see that, 1) in most cases, the values of cumulative wealth and cumulative risk obtained by the hybrid MFA-SOS algorithm are better than those obtained by the other algorithms; 2) in all cases, the mean values of the cumulative skewness

obtained by the MFA-SOS are the best; 3) the standard deviations obtained by the hybrid MFA-SOS algorithm are lower than those of the other algorithms in most cases, which indicates that the proposed algorithm has better stability.

Additionally, the results under different upper bounds are presented in Tables 7 and 8. We can also find that

Table 5 Performance comparisons in three periods under the cardinality constraint $m_t = 7$

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$t = 1$	Wealth	mean	1.177494	1.150616	1.175678	1.131715	1.140326	1.165122	1.150648	1.137133	1.131232
		std	0.001247	0.006832	0.001564	0.007763	0.011795	0.003103	0.017174	0.012104	0.009383
		max	1.180156	1.163820	1.177834	1.160731	1.161671	1.169713	1.167611	1.155722	1.149780
		min	1.173962	1.138233	1.172420	1.124324	1.116338	1.156444	1.101788	1.113835	1.116503
	Semi-entropy	mean	0.076599	0.067793	0.075840	0.071486	0.068114	0.070295	0.063626	0.072165	0.074311
		std	0.000943	0.004890	0.001115	0.002206	0.006038	0.004938	0.008842	0.005399	0.004637
		max	0.078712	0.079353	0.078341	0.075621	0.078406	0.077415	0.079247	0.081424	0.081208
		min	0.075051	0.058394	0.073571	0.067382	0.058876	0.061712	0.047073	0.061572	0.064719
	Skewness	mean	0.010211	0.004731	0.009684	0.004046	0.003838	0.006319	0.004474	0.004520	0.004504
		std	0.000319	0.001037	0.000348	0.000750	0.000798	0.001132	0.002072	0.001371	0.001422
		max	0.010865	0.007073	0.010333	0.006839	0.005285	0.007682	0.009140	0.007231	0.007767
		min	0.009756	0.003028	0.008971	0.003348	0.002568	0.004253	0.001879	0.002607	0.002303
$t = 2$	Wealth	mean	1.292614	1.287647	1.291820	1.251847	1.262704	1.306079	1.288534	1.260657	1.252859
		std	0.003221	0.016284	0.004628	0.008443	0.021625	0.014094	0.032083	0.019716	0.012529
		max	1.298400	1.315260	1.297870	1.267671	1.302368	1.325612	1.332512	1.284980	1.279058
		min	1.287083	1.264055	1.280726	1.236087	1.216532	1.279680	1.199694	1.218275	1.230085
	Semi-entropy	mean	0.133091	0.143120	0.133098	0.143924	0.135295	0.141388	0.136016	0.144366	0.148861
		std	0.001197	0.009528	0.001573	0.007410	0.007627	0.006727	0.008491	0.011066	0.005080
		max	0.134514	0.161149	0.135439	0.158031	0.147291	0.152288	0.151255	0.177410	0.157900
		min	0.130231	0.126581	0.130676	0.130123	0.117212	0.126193	0.114944	0.121057	0.138448
	Skewness	mean	0.015425	0.008845	0.014666	0.007821	0.006245	0.011830	0.009844	0.008471	0.008494
		std	0.000310	0.002667	0.000450	0.001163	0.001097	0.001450	0.002524	0.001847	0.001405
		max	0.016258	0.015210	0.015381	0.010812	0.008498	0.014048	0.015791	0.013253	0.012733
		min	0.014902	0.005233	0.013886	0.006048	0.004718	0.008852	0.004952	0.005318	0.006427
$t = 3$	Wealth	mean	1.423117	1.385675	1.415872	1.304184	1.344433	1.389867	1.357590	1.317741	1.313008
		std	0.006977	0.028352	0.010758	0.012430	0.035478	0.015443	0.056041	0.023832	0.014413
		max	1.432827	1.424023	1.426539	1.331583	1.400050	1.415492	1.443136	1.355314	1.338290
		min	1.407973	1.335833	1.388798	1.282736	1.270479	1.356361	1.225143	1.269825	1.281308
	Semi-entropy	mean	0.153886	0.181799	0.154719	0.187111	0.176168	0.168822	0.167899	0.184039	0.192484
		std	0.001401	0.012070	0.001432	0.009630	0.011270	0.007283	0.010024	0.012798	0.007539
		max	0.155649	0.204410	0.157281	0.205770	0.198100	0.180370	0.184663	0.216675	0.203018
		min	0.150391	0.160888	0.152746	0.170857	0.159916	0.152807	0.152123	0.157666	0.171355
	Skewness	mean	0.015655	0.008986	0.014838	0.007935	0.006368	0.011929	0.009947	0.008568	0.008630
		std	0.000358	0.002652	0.000460	0.001191	0.001124	0.001459	0.002523	0.001861	0.001417
		max	0.016444	0.015329	0.015557	0.010966	0.008596	0.014127	0.015850	0.013429	0.012922
		min	0.015006	0.005330	0.014022	0.006091	0.004776	0.008931	0.004920	0.005409	0.006549

Table 6 Performance comparisons in three periods under the cardinality constraint $m_t = 9$

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCOA	MOFOA
$t = 1$	Wealth	mean	1.176751	1.149562	1.174111	1.168356	1.128359	1.161457	1.159545	1.141804	1.132693
		std	0.002666	0.009609	0.005667	0.037101	0.010382	0.002107	0.018530	0.019637	0.016167
		max	1.180474	1.167436	1.178071	1.212959	1.142944	1.165430	1.187241	1.178292	1.152062
		min	1.171559	1.125086	1.152364	1.109395	1.106177	1.157504	1.114595	1.106771	1.076801
	Semi-entropy	mean	0.075677	0.071766	0.075031	0.063905	0.069739	0.071659	0.066721	0.074490	0.075332
		std	0.002766	0.005186	0.005321	0.008999	0.004321	0.003832	0.013829	0.007998	0.007241
		max	0.080616	0.079137	0.083478	0.080169	0.075639	0.078234	0.088973	0.087813	0.087053
		min	0.069324	0.061857	0.056548	0.049539	0.062301	0.064125	0.047285	0.059776	0.064098
	Skewness	mean	0.009989	0.005299	0.009347	0.007420	0.003224	0.006080	0.005934	0.005488	0.005155
		std	0.000759	0.001246	0.001693	0.003761	0.000777	0.000812	0.004207	0.002774	0.001903
		max	0.011139	0.007255	0.011658	0.014316	0.004893	0.007510	0.013405	0.012210	0.009039
		min	0.008434	0.003332	0.003280	0.002873	0.002270	0.004389	0.001228	0.002159	0.002297
$t = 2$	Wealth	mean	1.300445	1.282526	1.293502	1.289874	1.243877	1.293193	1.292458	1.265524	1.257329
		std	0.013743	0.018073	0.005398	0.026740	0.015284	0.014821	0.023554	0.018114	0.015743
		max	1.337444	1.317940	1.299033	1.339386	1.274726	1.319237	1.329985	1.293744	1.276498
		min	1.283289	1.242258	1.279684	1.245865	1.218597	1.263113	1.245875	1.229581	1.224961
	Semi-entropy	mean	0.135148	0.147062	0.135367	0.135894	0.137473	0.141484	0.135236	0.145824	0.152283
		std	0.002822	0.007268	0.003543	0.015028	0.007306	0.007674	0.009394	0.007762	0.007095
		max	0.142069	0.168782	0.147003	0.169163	0.150300	0.156333	0.150860	0.167041	0.164341
		min	0.130303	0.135252	0.128157	0.110332	0.122486	0.122861	0.117568	0.136024	0.137135
	Skewness	mean	0.015216	0.009094	0.014728	0.011947	0.005435	0.010736	0.010715	0.009653	0.010127
		std	0.000552	0.002173	0.000945	0.002747	0.001024	0.001415	0.003335	0.002246	0.002056
		max	0.016162	0.015399	0.016003	0.015728	0.008094	0.013250	0.015737	0.014280	0.014784
		min	0.014329	0.005747	0.011543	0.005683	0.003551	0.007111	0.003559	0.006968	0.006176
$t = 3$	Wealth	mean	1.432914	1.372203	1.421839	1.347065	1.317345	1.377845	1.372460	1.321220	1.325679
		std	0.018824	0.022469	0.008548	0.033341	0.025652	0.017459	0.055406	0.021589	0.022469
		max	1.479939	1.426290	1.431362	1.416541	1.364127	1.411541	1.446611	1.353706	1.377120
		min	1.409372	1.335864	1.402918	1.299206	1.270690	1.344449	1.262568	1.277302	1.282610
	Semi-entropy	mean	0.155596	0.189286	0.157104	0.182339	0.180700	0.171890	0.163851	0.187311	0.190971
		std	0.002621	0.009320	0.003414	0.016997	0.009463	0.008119	0.009517	0.012583	0.008297
		max	0.161405	0.216968	0.169218	0.224530	0.197080	0.189437	0.181607	0.212711	0.208458
		min	0.150349	0.175029	0.152735	0.156954	0.163886	0.152693	0.151866	0.165210	0.170432
	Skewness	mean	0.015402	0.009265	0.014914	0.012169	0.005577	0.010852	0.010820	0.009769	0.010272
		std	0.000549	0.002207	0.000947	0.002708	0.001001	0.001420	0.003318	0.002252	0.002059
		max	0.016356	0.015786	0.016192	0.015817	0.008258	0.013386	0.015937	0.014330	0.014953
		min	0.014529	0.005845	0.011724	0.005754	0.003806	0.007213	0.003877	0.007067	0.006343

the proposed MFA-SOS outperforms the other compared algorithms. From these tables, along with the dynamic adjustment of the portfolio, the satisfaction degree of the three objectives will increase. At each period, the proposed MFA-SOS obtains solutions with better performance than the other algorithms. Especially in the last period ($t =$

3), MFA-SOS algorithm has the best terminal wealth and skewness, and the smallest risk, which indicates that the strategies obtained by the proposed algorithm are more satisfactory to investors than the other algorithms.

Furthermore, four widely adopted metrics, the generation distance(GD) [76], the inverted generation distance (IGD)

Table 7 Performance comparisons in three periods under the upper bound $\mu = 0.3$

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$t = 1$	Wealth	mean	1.178802	1.155609	1.177293	1.142674	1.152936	1.164334	1.158060	1.151910	1.143543
		std	0.001101	0.004683	0.001417	0.003556	0.013779	0.002299	0.017347	0.009435	0.005991
		max	1.180440	1.165511	1.179619	1.147843	1.167420	1.167420	1.178961	1.165748	1.155857
		min	1.177018	1.148270	1.174441	1.134335	1.122340	1.157951	1.115738	1.130132	1.130783
	Semi-entropy	mean	0.076467	0.065177	0.075884	0.070609	0.065612	0.064711	0.067108	0.070448	0.069906
		std	0.000760	0.003926	0.001105	0.001824	0.004707	0.000809	0.010152	0.005501	0.004628
		max	0.077634	0.071230	0.077957	0.074575	0.075852	0.066259	0.081471	0.077823	0.077021
		min	0.075142	0.057739	0.073606	0.067908	0.057243	0.062740	0.047991	0.059547	0.058152
	Skewness	mean	0.010267	0.004878	0.009878	0.004721	0.004441	0.004984	0.005674	0.005373	0.004716
		std	0.000306	0.001189	0.000361	0.000381	0.001056	0.000209	0.002611	0.001394	0.000924
		max	0.010781	0.007762	0.010583	0.005635	0.006383	0.005274	0.009927	0.008041	0.006096
		min	0.009723	0.003282	0.009180	0.004085	0.001935	0.004355	0.001536	0.003241	0.002686
$t = 2$	Wealth	mean	1.294983	1.301302	1.292851	1.263944	1.280967	1.309002	1.299461	1.274962	1.270885
		std	0.003245	0.008900	0.003366	0.006503	0.016611	0.007055	0.030473	0.015997	0.006767
		max	1.302824	1.316718	1.299956	1.275954	1.303699	1.323145	1.344583	1.299608	1.284715
		min	1.291331	1.284567	1.286169	1.246594	1.242534	1.299077	1.207616	1.244940	1.257500
	Semi-entropy	mean	0.133174	0.139584	0.132712	0.140826	0.132423	0.138083	0.141041	0.139258	0.140721
		std	0.001315	0.006429	0.001443	0.004474	0.009524	0.005790	0.009763	0.007646	0.006577
		max	0.135306	0.155402	0.135532	0.149285	0.155510	0.153546	0.155697	0.155607	0.154378
		min	0.131234	0.126890	0.130461	0.130196	0.120492	0.132912	0.117937	0.125163	0.131025
	Skewness	mean	0.015470	0.009685	0.015016	0.008800	0.006905	0.010195	0.012298	0.009461	0.009034
		std	0.000413	0.002295	0.000415	0.000709	0.001474	0.001024	0.002669	0.001316	0.001064
		max	0.016410	0.014419	0.015856	0.010117	0.010775	0.012596	0.016332	0.012205	0.011108
		min	0.014792	0.005082	0.014251	0.006850	0.003828	0.009252	0.005378	0.006169	0.007576
$t = 3$	Wealth	mean	1.423102	1.410259	1.414055	1.321972	1.367032	1.378188	1.369627	1.337498	1.336989
		std	0.006450	0.014676	0.007329	0.010855	0.024694	0.011371	0.036193	0.029420	0.012188
		max	1.434017	1.429888	1.425585	1.343777	1.401216	1.397437	1.430018	1.389264	1.361974
		min	1.409977	1.375455	1.395390	1.304092	1.323586	1.364264	1.300720	1.292270	1.312504
	Semi-entropy	mean	0.153001	0.172726	0.153642	0.179753	0.168174	0.169191	0.172361	0.172778	0.177076
		std	0.001413	0.007336	0.001506	0.007447	0.010155	0.006472	0.014299	0.009148	0.008174
		max	0.155347	0.189749	0.156318	0.197660	0.192444	0.181119	0.206148	0.188645	0.191864
		min	0.150671	0.160315	0.151226	0.167369	0.151732	0.162058	0.152375	0.160100	0.164770
	Skewness	mean	0.015603	0.009816	0.015178	0.008924	0.007000	0.010268	0.012385	0.009539	0.009146
		std	0.000410	0.002274	0.000413	0.000742	0.001455	0.001015	0.002667	0.001322	0.001054
		max	0.016586	0.014432	0.016024	0.010313	0.010791	0.012635	0.016328	0.012292	0.011204
		min	0.014975	0.005276	0.014427	0.006944	0.003887	0.009348	0.005474	0.006230	0.007736

[76], the maximum spread (MS) [77] and the spacing (Spacing) [78], are used to evaluate the performance of the proposed MFA-SOS algorithm.

(1) Generation Distance (GD)

$$GD = \frac{\sqrt{\sum_{i=1}^n d_i^2}}{n}, \quad (45)$$

where n is the number of non-dominated solutions obtained by the algorithm, and d_i is the minimum distance between the i th solution and the real Pareto optimal solution set. If the value of GD is close to 0, the non-dominated solution set is close to the true Pareto optimal set. The GD metric reflects the approximate degree between the optimal solution set obtained by the algorithm and the real Pareto optimal solution set.

Table 8 Performance comparisons in three periods under the upper bound $\mu = 0.7$

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$t = 1$	Wealth	mean	1.178145	1.152816	1.177543	1.145016	1.156242	1.176752	1.154422	1.146839	1.145106
		std	0.001859	0.009628	0.001108	0.006020	0.012398	0.005047	0.031772	0.014401	0.00560
		max	1.182295	1.174134	1.179619	1.155369	1.172476	1.179977	1.181900	1.164829	1.153078
		min	1.173068	1.129029	1.175911	1.126829	1.134793	1.161709	1.059628	1.105526	1.131862
	Semi-entropy	mean	0.075654	0.065527	0.076199	0.071158	0.065663	0.077145	0.065319	0.069431	0.072089
		std	0.000818	0.004929	0.000963	0.003157	0.007262	0.007743	0.012946	0.005609	0.002531
		max	0.076979	0.078529	0.078905	0.076088	0.085197	0.084739	0.089512	0.081179	0.076164
		min	0.074101	0.055209	0.074989	0.059218	0.050741	0.056709	0.044079	0.059484	0.065594
	Skewness	mean	0.010135	0.004976	0.009997	0.005257	0.004691	0.009496	0.005431	0.004744	0.005171
		std	0.000350	0.001442	0.000356	0.000703	0.001548	0.002238	0.003655	0.001338	0.000628
		max	0.011194	0.009837	0.010992	0.006881	0.007881	0.011844	0.013857	0.007150	0.006302
		min	0.009636	0.003075	0.009605	0.003541	0.001981	0.003557	0.000408	0.002289	0.003845
$t = 2$	Wealth	mean	1.294235	1.293830	1.291814	1.268950	1.289421	1.293272	1.287241	1.273835	1.266658
		std	0.003996	0.017482	0.003808	0.009248	0.023095	0.016054	0.042599	0.018661	0.009267
		max	1.299397	1.323284	1.297493	1.284487	1.330833	1.319798	1.347968	1.317981	1.282511
		min	1.287491	1.253566	1.283346	1.245931	1.245773	1.269718	1.175853	1.233442	1.248921
	Semi-entropy	mean	0.132331	0.139565	0.132078	0.143587	0.131819	0.133230	0.135232	0.140680	0.141306
		std	0.001563	0.009301	0.001680	0.005089	0.010371	0.004352	0.013812	0.009333	0.004859
		max	0.134921	0.162922	0.134464	0.150447	0.164096	0.141904	0.160594	0.153421	0.149521
		min	0.130496	0.122059	0.128976	0.129492	0.117212	0.128033	0.106884	0.122466	0.131406
	Skewness	mean	0.015420	0.010091	0.014886	0.009921	0.007138	0.012830	0.011218	0.009260	0.009075
		std	0.000525	0.002776	0.000478	0.000953	0.001675	0.000847	0.003430	0.001923	0.000836
		max	0.016301	0.014936	0.015825	0.011905	0.010114	0.014658	0.015766	0.013433	0.010926
		min	0.014552	0.004758	0.013866	0.008264	0.004718	0.011238	0.004686	0.006326	0.007922
$t = 3$	Wealth	mean	1.422859	1.394747	1.411082	1.331811	1.384404	1.399303	1.360614	1.340015	1.336478
		std	0.008022	0.022618	0.008917	0.008724	0.028754	0.016875	0.057394	0.022754	0.014979
		max	1.432135	1.431169	1.429485	1.342200	1.423231	1.430254	1.426105	1.389451	1.368843
		min	1.407361	1.342531	1.395051	1.309357	1.319364	1.378720	1.190502	1.297834	1.305630
	Semi-entropy	mean	0.152707	0.174727	0.153097	0.183954	0.166747	0.156015	0.168895	0.176951	0.178778
		std	0.001776	0.008394	0.001251	0.006136	0.012925	0.003984	0.010946	0.011162	0.006369
		max	0.155246	0.195508	0.155547	0.193439	0.198888	0.163364	0.188236	0.195161	0.188540
		min	0.150039	0.160039	0.150867	0.170587	0.151008	0.151104	0.151692	0.158786	0.165789
	Skewness	mean	0.015696	0.010223	0.015094	0.010063	0.007274	0.012945	0.011347	0.009367	0.009199
		std	0.000579	0.002804	0.000415	0.000962	0.001675	0.000847	0.003434	0.001911	0.000854
		max	0.016814	0.015148	0.015980	0.012122	0.010296	0.014794	0.016236	0.013501	0.011097
		min	0.014819	0.004951	0.014118	0.008489	0.004776	0.011350	0.004675	0.006564	0.008005

(2) Inverted Generation Distance (IGD)

$$IGD = \frac{\sqrt{\sum_{i=1}^m d_i^2}}{m}, \quad (46)$$

where m is the number of solutions in the real Pareto front, and d_i is the Euclidean distance among each of the solutions and the nearest member from the non-dominated solution

set. If the IGD is small, the solution set obtained by the algorithm is close to the real Pareto optimal set.

(3) Maximum Spread (MS)

$$MS = \sqrt{\frac{1}{k} \sum_{l=1}^k \delta_l}, \quad \delta_l = \left(\frac{\min(f_l^{\max}, F_l^{\max}) - \max(f_l^{\min}, F_l^{\min})}{F_l^{\max} - F_l^{\min}} \right)^2 \quad (47)$$

Table 9 Performance comparisons in four metrics under different values of m_t

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$m_t = 5$	GD	mean	0.006022	0.054531	0.007537	0.039271	0.029834	0.015341	0.078309	0.050773	0.025872
		std	0.000764	0.028448	0.000809	0.003530	0.005598	0.001464	0.044620	0.030344	0.018034
		max	0.008365	0.090320	0.009658	0.046857	0.043687	0.019323	0.142332	0.120059	0.091571
		min	0.005060	0.023083	0.006624	0.033370	0.023065	0.013148	0.009176	0.022640	0.014881
	IGD	mean	0.005323	0.040978	0.005801	0.044047	0.052839	0.036788	0.043589	0.051545	0.042766
		std	0.000608	0.005956	0.000687	0.002204	0.004990	0.004989	0.017119	0.006332	0.002346
		max	0.006406	0.056247	0.007648	0.047919	0.060061	0.045225	0.082371	0.060915	0.050873
		min	0.004468	0.030077	0.004916	0.039801	0.042623	0.028934	0.016239	0.036903	0.039754
	MS	mean	0.892635	0.484550	0.848319	0.393132	0.096234	0.141282	0.401890	0.269747	0.227412
		std	0.029885	0.088849	0.032653	0.032155	0.052831	0.063555	0.194747	0.124781	0.117679
		max	0.962041	0.606178	0.911852	0.467082	0.237842	0.269425	0.761736	0.514352	0.537438
		min	0.850443	0.333464	0.789802	0.329939	0.032530	0.053183	0.109200	0.126957	0.043407
	Spacing	mean	0.023823	0.047630	0.041732	0.059169	0.041178	0.028840	0.031642	0.042254	0.046809
		std	0.001530	0.021142	0.004732	0.017187	0.019323	0.004191	0.009923	0.019131	0.016721
		max	0.026360	0.090355	0.053497	0.109042	0.085114	0.036981	0.057637	0.100868	0.077414
		min	0.020936	0.019954	0.034928	0.032630	0.009180	0.021577	0.017438	0.021294	0.023528
$m_t = 7$	GD	mean	0.005029	0.036152	0.007163	0.051368	0.038683	0.018874	0.016020	0.021454	0.018489
		std	0.000562	0.005148	0.002371	0.003917	0.011560	0.001771	0.011283	0.004281	0.001596
		max	0.006511	0.046902	0.013904	0.059785	0.059524	0.023580	0.049429	0.030078	0.020708
		min	0.004271	0.027909	0.005421	0.046252	0.017382	0.015880	0.005976	0.014352	0.014186
	IGD	mean	0.003847	0.053328	0.005305	0.063179	0.066358	0.042529	0.040224	0.044094	0.046885
		std	0.000465	0.006882	0.000844	0.003869	0.006509	0.004926	0.014155	0.005323	0.001781
		max	0.004784	0.065622	0.007099	0.069295	0.081919	0.050260	0.059339	0.052285	0.051868
		min	0.003097	0.042797	0.004081	0.053699	0.056939	0.033823	0.017545	0.034268	0.044472
	MS	mean	0.936101	0.255120	0.849842	0.206045	0.019169	0.055894	0.221977	0.155735	0.110226
		std	0.031348	0.042206	0.031013	0.026058	0.012713	0.030889	0.140965	0.205386	0.052814
		max	0.996220	0.341146	0.907033	0.266354	0.055462	0.130948	0.530223	0.564689	0.323955
		min	0.881548	0.202806	0.794047	0.160656	0.002635	0.011082	0.067619	0.023673	0.067077
	Spacing	mean	0.016833	0.052200	0.037251	0.068111	0.051884	0.030714	0.042563	0.045305	0.031091
		std	0.001308	0.025507	0.019321	0.019682	0.025942	0.018248	0.033529	0.023388	0.007375
		max	0.018791	0.102406	0.086571	0.101834	0.113119	0.087743	0.132642	0.087024	0.045231
		min	0.014474	0.015868	0.023242	0.031997	0.016553	0.011235	0.014945	0.015239	0.020666
$m_t = 9$	GD	mean	0.005751	0.058299	0.007641	0.029393	0.035952	0.018932	0.065281	0.099645	0.083083
		std	0.000563	0.028690	0.002673	0.009863	0.006029	0.001215	0.045217	0.044703	0.032869
		max	0.007634	0.109529	0.015077	0.052957	0.045981	0.021559	0.179500	0.172119	0.160467
		min	0.005204	0.022484	0.005311	0.012950	0.024934	0.016824	0.009504	0.035263	0.037004
	IGD	mean	0.010449	0.041324	0.012026	0.034572	0.056071	0.035745	0.039353	0.048282	0.065174
		std	0.000380	0.002827	0.002085	0.007540	0.006033	0.002428	0.012933	0.005491	0.001467
		max	0.010957	0.045649	0.017791	0.049667	0.071208	0.039805	0.068739	0.055658	0.067934
		min	0.009142	0.036610	0.010241	0.020527	0.046437	0.031024	0.016603	0.034885	0.061794
	MS	mean	0.444118	0.308277	0.394557	0.229205	0.026418	0.076416	0.240940	0.303624	0.265459
		std	0.029088	0.142633	0.035950	0.103199	0.017446	0.034279	0.106133	0.132732	0.143068
		max	0.506110	0.472159	0.469042	0.440333	0.066404	0.162589	0.435788	0.402196	0.355842
		min	0.400214	0.106244	0.336641	0.096881	0.003063	0.031865	0.066194	0.030293	0.016020

Table 9 (continued)

		MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
Spacing	mean	0.024372	0.047122	0.033293	0.044845	0.042364	0.031408	0.033413	0.042700	0.056497
	std	0.001527	0.017982	0.010692	0.026042	0.024369	0.005894	0.017021	0.019486	0.014056
	max	0.027322	0.076847	0.058426	0.110155	0.086438	0.049167	0.067423	0.081243	0.071065
	min	0.019907	0.018475	0.021803	0.014821	0.016612	0.022091	0.012664	0.005426	0.027296

where $f_l^{\max}$ and $f_l^{\min}$ are the maximum and minimum values of the l th objective function obtained by the algorithm, and $F_l^{\max}$ and $F_l^{\min}$ are the maximum and

minimum values of the l th objective function for the real Pareto frontier. k is the number of objective functions. $MS = 1$, indicates that the true Pareto frontier is completely

Table 10 Performance comparisons in four metrics under different values of μ

			MFA-SOS	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
u=0.3	GD	mean	0.005604	0.027827	0.006432	0.039524	0.028895	0.022428	0.064946	0.069008	0.032836
		std	0.000380	0.005107	0.000618	0.003292	0.006407	0.001179	0.041598	0.038174	0.026794
		max	0.006126	0.037496	0.007351	0.047676	0.043437	0.024754	0.138095	0.138680	0.086532
		min	0.004833	0.019880	0.005429	0.035676	0.017850	0.020491	0.011275	0.023910	0.010544
	IGD	mean	0.005729	0.038792	0.006399	0.046521	0.054144	0.041465	0.034358	0.048243	0.036650
		std	0.000819	0.007269	0.000523	0.001933	0.005390	0.002659	0.018253	0.007109	0.003213
		max	0.006656	0.057793	0.007334	0.049686	0.063495	0.045350	0.083545	0.063728	0.043105
		min	0.004165	0.029472	0.005639	0.042866	0.046792	0.034140	0.011345	0.034883	0.031180
	MS	mean	0.909876	0.409252	0.842230	0.350173	0.075950	0.145590	0.443107	0.349480	0.311477
		std	0.038224	0.080763	0.039998	0.022670	0.041446	0.049634	0.208170	0.140641	0.118283
		max	0.973688	0.622495	0.920490	0.394362	0.163197	0.226767	0.829920	0.568508	0.548967
		min	0.829617	0.318512	0.774626	0.310476	0.029936	0.067342	0.051266	0.060616	0.102142
	Spacing	mean	0.019759	0.049192	0.025797	0.065627	0.040219	0.030658	0.029966	0.046940	0.043775
		std	0.001622	0.018382	0.007430	0.018054	0.026991	0.008603	0.016630	0.018692	0.011427
		max	0.022540	0.085664	0.040618	0.103655	0.098952	0.054021	0.079104	0.079116	0.064813
		min	0.017076	0.022127	0.015962	0.038921	0.014940	0.017585	0.006780	0.016083	0.020776
	GD	mean	0.006823	0.031013	0.007641	0.038191	0.022974	0.010744	0.095110	0.079813	0.026050
		std	0.000590	0.013583	0.000463	0.002634	0.007337	0.001789	0.049013	0.036546	0.021664
		max	0.007629	0.087444	0.008296	0.043931	0.044043	0.014076	0.186867	0.131511	0.089535
		min	0.005667	0.023451	0.006703	0.032638	0.011484	0.008021	0.011497	0.022138	0.014558
	IGD	mean	0.005661	0.039921	0.006489	0.044304	0.051387	0.037343	0.049626	0.050575	0.042311
		std	0.001250	0.007459	0.001119	0.001725	0.006624	0.005816	0.019827	0.007049	0.003334
		max	0.007849	0.056968	0.008128	0.047711	0.062111	0.048916	0.077165	0.062115	0.048239
		min	0.004211	0.026093	0.004943	0.041025	0.040404	0.029640	0.016349	0.035845	0.035629
	MS	mean	0.887890	0.405652	0.820432	0.408503	0.087366	0.127058	0.364081	0.384198	0.240712
		std	0.028221	0.063251	0.042811	0.053856	0.051621	0.056301	0.175345	0.140495	0.084399
		max	0.925672	0.563297	0.864793	0.555961	0.227420	0.207313	0.691141	0.533468	0.419223
		min	0.836508	0.321430	0.706475	0.342017	0.014171	0.030865	0.152347	0.088703	0.104961
	Spacing	mean	0.023522	0.045697	0.044382	0.071688	0.043693	0.030414	0.032383	0.049721	0.037180
		std	0.001815	0.017448	0.002772	0.016002	0.031966	0.009890	0.013651	0.014869	0.014809
		max	0.026547	0.084127	0.049110	0.103468	0.152074	0.051384	0.063017	0.082133	0.076963
		min	0.019813	0.016880	0.039319	0.031034	0.010435	0.015984	0.008636	0.025101	0.015086

covered by the solutions obtained by the algorithm. The MS is the metric representing the coverage of the Pareto frontier obtained by the algorithm to the true Pareto front.

(4) Spacing

$$Spacing = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\bar{d}_i - d_i)^2}, \quad (48)$$

where n is the number of non-dominated solutions obtained by the algorithm, d_i is the distance between the i th non-dominated solution corresponding to the target vector and the nearest target vector, and $\bar{d}_i$ is the mean of the d_i . A

smaller Spacing value demonstrates that the solutions are more evenly distributed.

To compare the performance on the metrics, similar to [37], we combine all the results obtained by the nine algorithms. According to the Pareto dominance relationship, an approximate Pareto front is obtained by selecting a certain number of non-dominated solutions from all the results. Then, the approximate Pareto front will be regarded as the real Pareto front. Two cases, namely $m_t = 5, 7, 9$ and $\mu = 0.3, 0.7$, are respectively considered, and the results of the above four performance metrics obtained by the nine algorithms are listed in Tables 9 and 10. From Table 9, we observe that MFA-SOS outperforms the other

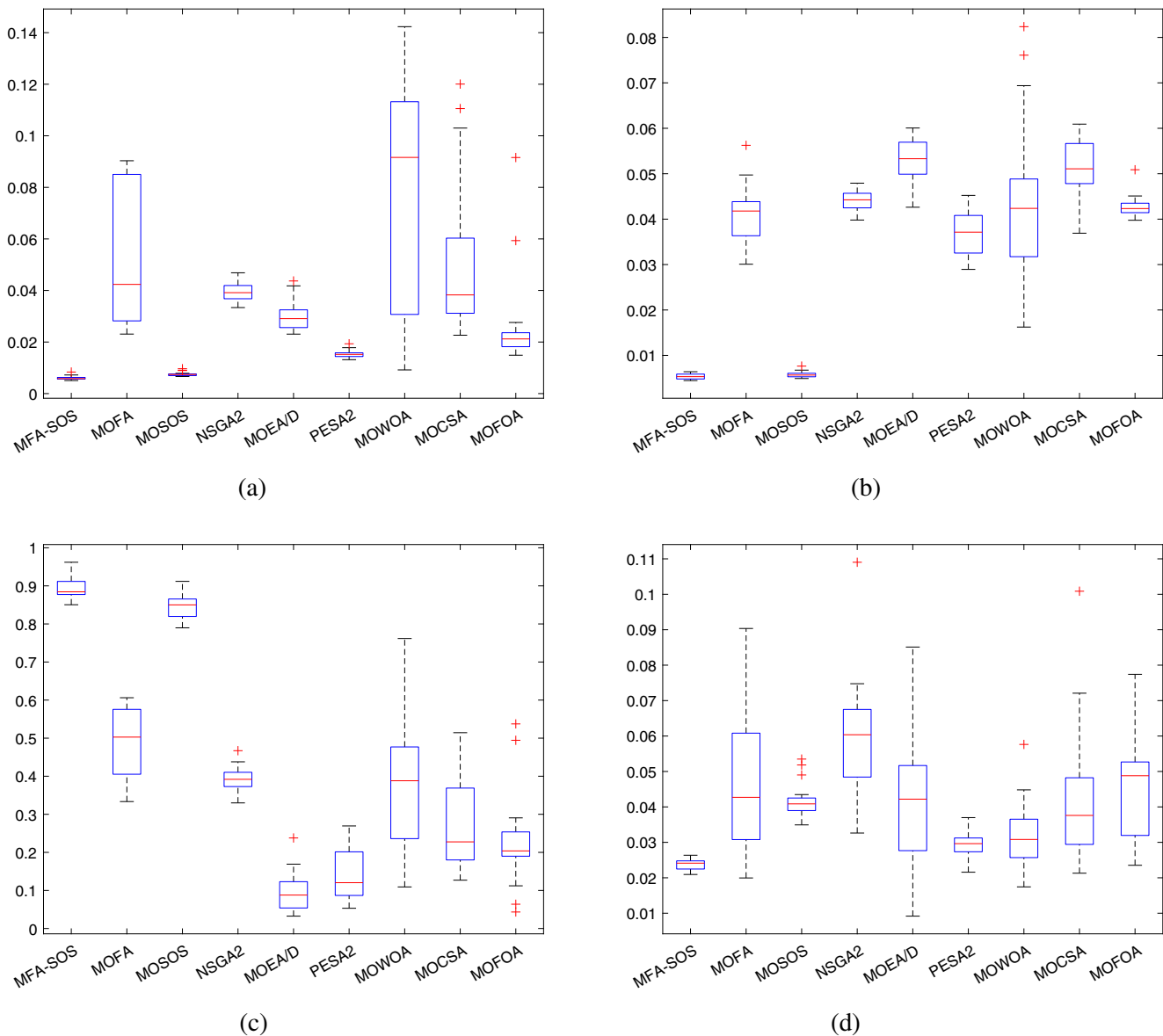


Fig. 1 Boxplots of performance measure metrics in the case of $mt = 5$. **a** GD. **b** IGD. **c** MS. **d** Spacing

eight algorithms in terms of GD and IGD in all cases, which shows that the solutions obtained by the hybrid MFA-SOS algorithm are close to the real Pareto front. Compared with other algorithms, the MFA-SOS algorithm obtains the best mean of MS and the smallest mean of Spacing. And the other performance indicators, (i.e., std, max and min) also outperform the compared algorithms in most cases, which means that the obtained solutions of the proposed algorithm are more evenly distributed and cover a wide range. Table 10 shows the performance metrics in the cases of $\mu = 0.3$ and $\mu = 0.7$. We can see that, the MFA-SOS is superior to the other eight algorithms according to the results of the mean values of GD, IGD, MS and Spacing.

Although MOSOS has a smaller std than MFA-SOS in some cases, it concentrates on the stability of IGD and shows negligible advantages in accuracy and distribution range of solutions. Overall, the hybrid MFA-SOS algorithm is more effective in solving the proposed model than the other algorithms.

In addition to the statistics listed in tabular form, Figs. 1, 2 and 3 visually show the boxplots of the four metrics for the nine algorithms in the cases of $m_t = 5$, $m_t = 7$ and $m_t = 9$. The results of the four performance metrics are shown in Figs. 1–3, where GD is in the upper left corner, IGD is in the upper right corner, MS is in the lower left corner, and Spacing is in the lower right corner

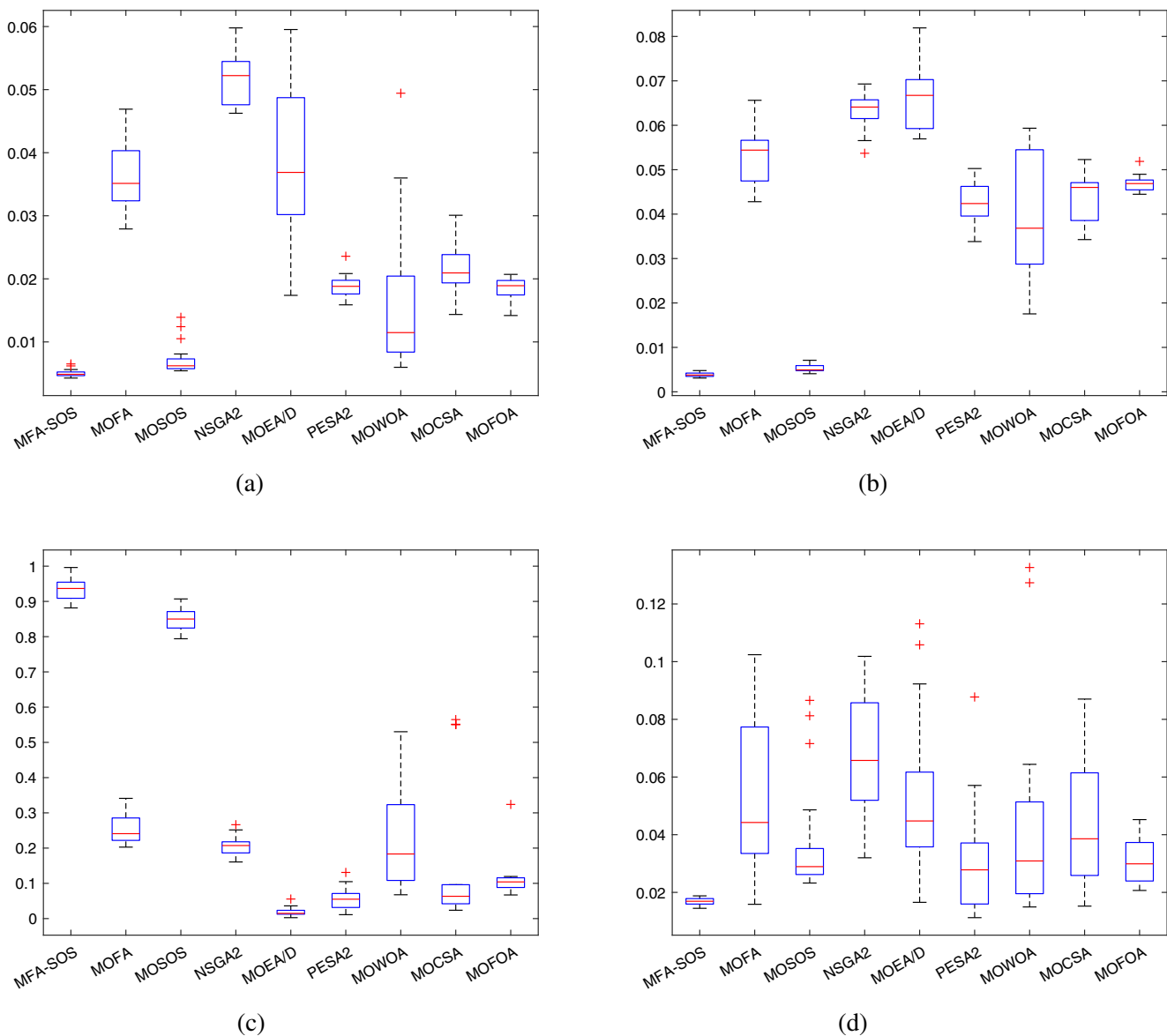


Fig. 2 Boxplots of performance measure metrics in the case of $m_t = 7$. **a** GD. **b** IGD. **c** MS. **d** Spacing

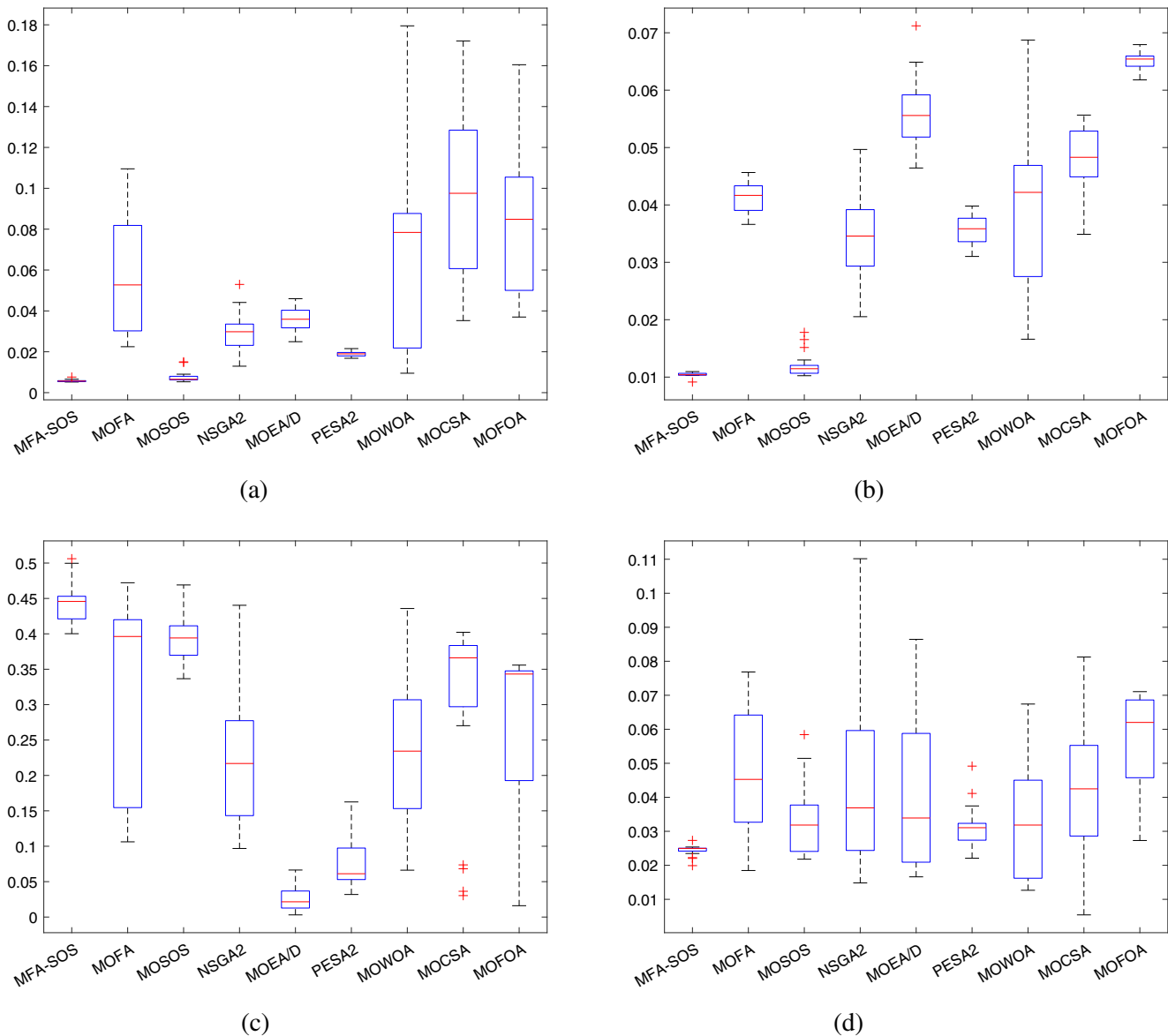


Fig. 3 Boxplots of performance measure metrics in the case of $mt = 9$. **a** GD. **b** IGD. **c** MS. **d** Spacing

corner. In Fig. 1, the box plot representing MFA-SOS is flat, which indicates that the standard deviation of the results obtained by MFA-SOS are more stable and have a smaller standard deviation. From Fig. 2, we find that the box plots representing the values of GD and IGD for MFA-SOS are at the bottom, which shows that its values are smaller than those of the other algorithms, and the solutions obtained by MFA-SOS are closer to the real Pareto front. For MS, the box diagram representing MFA-SOS is at the top, indicating that it has a larger solution coverage than the other compared algorithms. Moreover, it can be seen from Fig. 2 that the standard deviations of GD, IGD and Spacing for MFA-SOS are smaller than those of the other eight algorithms.

Similar to Figs. 1 and 2, the same conclusions can also be drawn from Fig. 3. In general, the boxplots reconfirm the superiority of the hybrid MFA-SOS algorithm in an intuitive form.

To confirm the above conclusions, a non-parametric test, the Wilcoxon rank sum test, is applied to the proposed algorithm and the compared algorithms. We perform the Wilcoxon rank sum test between MFA-SOS and each comparison algorithm (MOFA, MOSOS, NSGA-II, MOEA/D, PESA-II, MOWOA, MOCSA and MOFOA) on the results of the GD, IGD, MS and Spacing metrics obtained for 20 independent runs, with a significance level $\alpha = 0.05$. According to the p -value and the given significance level

Table 11 The p -values of the Wilcoxon rank sum test results under different values of m_t

	MFA-SOS vs.	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$m_t = 5$	GD	6.77647E-08	3.06257E-06	6.77647E-08	6.77647E-08	6.77647E-08	6.76692E-08	6.76692E-08	6.77647E-08
	IGD	6.77647E-08	3.37010E-02	6.77647E-08	6.77647E-08	6.77647E-08	6.76692E-08	6.76692E-08	6.77647E-08
	MS	6.77647E-08	1.79113E-04	6.77647E-08	6.77647E-08	6.77647E-08	6.76692E-08	6.76692E-08	6.77647E-08
	Spacing	5.88650E-05	6.77647E-08	6.77647E-08	1.47945E-03	3.74514E-04	1.11414E-03	3.28470E-05	7.92910E-07
$m_t = 7$	GD	6.79562E-08	2.06160E-06	6.79562E-08	6.79562E-08	6.79562E-08	1.22796E-07	6.74785E-08	6.78604E-08
	IGD	6.79562E-08	3.93881E-07	6.79562E-08	6.79562E-08	6.79562E-08	6.75738E-08	6.74785E-08	6.78604E-08
	MS	6.79562E-08	1.23464E-07	6.79562E-08	6.79562E-08	6.79562E-08	6.72883E-08	6.74785E-08	6.79562E-08
	Spacing	6.01480E-07	6.79562E-08	6.79562E-08	3.93881E-07	1.05812E-02	6.58961E-05	2.58468E-05	6.78604E-08
$m_t = 9$	GD	6.56899E-08	1.99729E-05	6.56899E-08	6.55969E-08	6.56899E-08	6.56899E-08	6.56899E-08	6.56899E-08
	IGD	6.56899E-08	1.27156E-04	6.56899E-08	6.55969E-08	6.56899E-08	6.56899E-08	6.56899E-08	6.56899E-08
	MS	2.43101E-04	5.78366E-05	5.07515E-07	6.55969E-08	6.56899E-08	5.84198E-07	8.87270E-08	6.56899E-08
	Spacing	8.14049E-05	3.82967E-02	8.28186E-03	2.92508E-02	1.53362E-06	3.58393E-02	5.50727E-03	7.63716E-08

, the following hypotheses for GD, IGD, MS and Spacing metrics are defined:

- H_0 : $p\text{-value} > \alpha$, which indicates that the GD, IGD, MS and Spacing of MFA-SOS are similar to those of other algorithms.
- H_1 : $p\text{-value} \leq \alpha$, which indicates that the GD, IGD and Spacing of MFA-SOS are smaller than those of the other algorithms, and the MS of MFA-SOS is larger than those of the other algorithms.

Tables 11 and 12 list the p -values of Wilcoxon rank sum test between the MFA-SOS and each comparison algorithm in terms of GD, IGD, MS and Spacing. In Tables 11 and 12, $p\text{-value} < 0.05$ indicates that the null hypothesis (H_0) is rejected. From Tables 11 and 12, we can see that, the p -values are less than 0.05 in all cases and MFA-SOS performs significantly different from the compared algorithms in terms of the comparison metrics. In summary,

MFA-SOS can achieve a significantly diverse set of non-dominated solutions than those obtained by the comparison algorithms.

6 Conclusions

In this paper, we discuss an uncertain multiobjective multiperiod portfolio selection problem. We present an uncertain multiobjective multiperiod mean-semientropy-skewness portfolio optimization model, in which several real-world constraints such as cardinality and liquidity are considered simultaneously. Then, a novel hybrid MFA-SOS algorithm is designed to solve the corresponding portfolio selection model. Finally, a numerical example is given to illustrate the validity of the proposed approach. In future work, we may extend the proposed model to a hybrid portfolio selection problem with the simultaneous

Table 12 The p -values of the Wilcoxon rank sum test results under different values of μ

	MFA-SOS vs.	MOFA	MOSOS	NSGA-II	MOEA/D	PESA-II	MOWOA	MOCSA	MOFOA
$\mu = 0.3$	GD	6.79562E-08	1.03114E-04	6.79562E-08	6.79562E-08	6.70985E-08	6.79562E-08	6.79562E-08	6.79562E-08
	IGD	6.79562E-08	3.59786E-02	6.79562E-08	6.79562E-08	6.70985E-08	6.79562E-08	6.79562E-08	6.79562E-08
	MS	6.79562E-08	1.79440E-05	6.79562E-08	6.79562E-08	6.70985E-08	7.89803E-08	6.79562E-08	6.79562E-08
	Spacing	9.17277E-08	1.23447E-02	6.79562E-08	1.48098E-03	1.55979E-06	4.11236E-02	1.20089E-06	1.65708E-07
$\mu = 0.7$	GD	6.79562E-08	1.03734E-04	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08
	IGD	6.79562E-08	1.43638E-02	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08
	MS	6.79562E-08	3.98736E-06	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08	6.79562E-08
	Spacing	5.25495E-05	6.79562E-08	6.79562E-08	2.79860E-03	1.43638E-02	6.55719E-03	1.23464E-07	2.13926E-03

presence of random and uncertain returns. Furthermore, the hybrid MFA-SOS algorithm proposed in this paper can also be applied to other multi-objective optimization problems, such as vehicle scheduling and shortest path problems. Finally, we will apply machine learning techniques [79, 80] for stock market predictions, and then integrate the asset preselection with the portfolio selection model.

Acknowledgements This research was supported by the Young Academic Innovation Team of Capital University of Economics and Business of China (No. QNTD202002), the Humanity and Social Science Foundation of Ministry of Education of China (No. 19YJAZH005), and the special fund of basic scientific research business fees of Beijing Municipal University of Capital University of Economics and Business (No. XRZ2020016).

References

- Markowitz H (1952) Portfolio selection. *J Finance* 7(1):77–91
- Sharpe WF (1964) Capital asset prices: a theory of market equilibrium under conditions of risk. *J Finance* 19(3):425–442
- Atsushi Y (1996) The mean-variance approach to portfolio optimization subject to transaction costs. *J Oper Res Soc Jpn* 19(1):425–442
- Markowitz H (1959) Portfolio selection: efficient diversification of investments. Wiley, New York
- Konno H, Yamazaki H (1991) Mean-absolute deviation portfolio optimization model and its applications to Tokyo stock market. *Manage Sci* 37(5):519–531
- Jorion P (1996) Risk²: Measuring the risk in value at risk. *Financial Anal J* 52(6):47–56
- Rockafellar RT, Uryasev S (2000) Optimization of conditional value-at-risk. *J Risk* 2(3):1071–1074
- Philippatos GC, Wilson CJ (1972) Entropy, market risk, and the selection of efficient portfolios. *Appl Econ* 4(3):209–220
- Nawrocki DN, Harding WH (1986) State-value weighted entropy as a measure of investment risk. *Appl Econ* 18(4):411–419
- Simonelli MR (2005) Indeterminacy in portfolio selection. *Eur J Oper Res* 163(1):170–176
- Zhou JD, Li X, Pedrycz W (2016) Mean-semi-entropy models of fuzzy portfolio selection. *IEEE Trans Fuzzy Syst* 24(6):1627–1636
- Chen W, Wang Y, Mehlaawat MK (2018) A hybrid FA-SA algorithm for fuzzy portfolio selection with transaction costs. *Ann Oper Res* 269:129–147
- Mansour N, Cherif MS, Abdelfattah W (2019) Multi-objective imprecise programming for financial portfolio selection with fuzzy returns. *Expert Syst Appl* 138:112810
- zhou J, Li X (2020) Mean-semi-entropy portfolio adjusting model with transaction costs. *J Data Inf Manag* 2:121–130
- Zadeh LA (1965) Fuzzy sets. *Inf Control* 8(3):338–353
- Liu BD (2007) Uncertainty theory, 2nd edn. Springer, Berlin
- Qin ZF (2015) Mean-variance model for portfolio optimization problem in the simultaneous presence of random and uncertain returns. *Eur J Oper Res* 245(2):480–488
- Liu BD (2010) Uncertainty theory: a branch of mathematics for modeling human uncertainty. Springer, Berlin
- Zhai J, Bai M (2018) Mean-risk model for uncertain portfolio selection with background risk. *J Comput Appl Math* 330:59–69
- Chen W, Wang Y, Gupta P, Mehlaawat MK (2018) A novel hybrid heuristic algorithm for a new uncertain mean-variance-risk-skewness portfolio selection model with real constraints. *Appl Intell* 48(9):2996–3018
- Wang XT, Huang XX (2019) A risk index to model uncertain portfolio investment with options. *Econ Model* 80:284–293
- Li B, Sun YF, Aw G, Teo KL (2019) Uncertain portfolio optimization problem under a minimax risk measure. *Appl Math Model* 76:274–281
- Huang XX, Yang TT (2020) How does background risk affect portfolio choice: an analysis based on uncertain mean-variance model with background risk. *J Bank Financ* 111:105726
- Huang XX, Qiao L (2012) A risk index model for multi-period uncertain portfolio selection. *Inf Sci* 217:108–116
- Li B, Zhu Y, Sun Y, Aw G, Teo KL (2018) Multi-period portfolio selection problem under uncertain environment with bankruptcy constraint. *Appl Math Model* 56:539–550
- Chen W, Li DD, Liu YJ (2018) A novel hybrid ICA-FA algorithm for multiperiod uncertain portfolio optimization model based on multiple criteria. *IEEE Trans Fuzzy Syst* 27(5):1023–1036
- Zhang P (2019) Multiperiod mean absolute deviation uncertain portfolio selection with real constraints. *Soft Comput* 23(13):5081–5098
- Chen W, Li DD, Lu S, Liu WY (2019) Multi-period mean-semivariance portfolio optimization based on uncertain measure. *Soft Comput* 23(15):6231–6247
- Chang TJ, Meade N, Beasley JE et al (2000) Heuristics for cardinality constrained portfolio optimisation. *Comput Oper Res* 27:1271–1302
- Deng GF, Lin WT, Lo CC (2012) Markowitz-based portfolio selection with cardinality constraints using improved particle swarm optimization. *Expert Syst Appl* 39(4):4558–4566
- Gong DW, Han YY, Sun JY (2018) A novel hybrid multi-objective artificial bee colony algorithm for blocking lot-streaming flow shop scheduling problems. *Knowl Based Syst* 148:115–130
- Gong DW, Sun J, Ji XF (2013) Evolutionary algorithms with preference polyhedron for interval multi-objective optimization problems. *Inf Sci* 233:141–161
- Gong DW, Xu B, Zhang Y, Guo YN, Yang SX (2020) A similarity-based cooperative co-evolutionary algorithm for dynamic interval multiobjective optimization problems. *IEEE Trans Evol Comput* 24(1):142–156
- Xu B, Gong DW, Guo YN, Rong M (2018) Environment sensitivity-based cooperative co-evolutionary algorithms for dynamic multi-objective optimization. *IEEE/ACM Trans Comput Biol Bioinform* 15(6):1877–1890
- Zhang Y, Gong DW, Ding ZH (2011) Handling multi-objective optimization problems with a multi-swarm cooperative particle swarm optimizer. *Expert Syst Appl* 38(11):13933–13941
- Wang B, Li Y, Wang S, Watada J (2018) A multi-objective portfolio selection model with fuzzy value-at-risk ratio. *IEEE Trans Fuzzy Syst* 26(6):3673–3687
- Chen W, Xu W (2019) A hybrid multiobjective bat algorithm for fuzzy portfolio optimization with real-world constraints. *Int J Fuzzy Syst* 21(1):291–307
- Silva YLT, Herthel AB, Subramanian A (2019) A multi-objective evolutionary algorithm for a class of mean-variance portfolio selection problems. *Expert Syst Appl* 133:225–241
- Kaucic M (2019) Equity portfolio management with cardinality constraints and risk parity control using multi-objective particle swarm optimization. *Comput Oper Res* 109:300–316
- Liagkouras K (2019) A new three-dimensional encoding multi-objective evolutionary algorithm with application to the portfolio optimization problem. *Knowl Based Syst* 163:186–203
- Yang XS (2008) Nature-inspired Metaheuristic algorithms. Luniver Press, UK
- Fan BB, Yang WW, Zhang ZF (2019) Solving the two-stage hybrid flow shop scheduling problem based on mutant

- firefly algorithm. *J Ambient Intell Humaniz Comput* 10(3):979–990
43. Wang CH, Nguyen TT, Pan JS, Dao TK (2017) An optimization approach for potential power generator outputs based on parallelized firefly algorithm. In: *Proceeding of the twelfth international conference on intelligent information hiding and multimedia signal processing*, pp 297–306
44. Yang XS (2013) Multiobjective firefly algorithm for continuous optimization. *Eng Comput* 29(2):175–184
45. Marichelvam MK, Prabaharan T, Yang XS (2014) A discrete firefly algorithm for the multi-objective hybrid flowshop scheduling problems. *IEEE Trans Evol Comput* 18(2):301–305
46. Karthikeyan S, Asokan P, Nickolas S, Page T (2015) A hybrid discrete firefly algorithm for solving Multi-Objective flexible job shop scheduling problems. *Int J Bio-Inspir Com* 7(6):386–401
47. Hidalgo-Paniagua A, Vega-rodríguez MA, Ferruz J, Pavón N (2017) Solving the multi-objective path planning problem in mobile robotics with a firefly-based approach. *Soft Comput* 21:949–964
48. Bozorg-Haddad O, Garousi-Nejad I, Loáiciga HA (2017) Extended multi-objective firefly algorithm for hydropower energy generation. *J Hydroinformatics* 19(5):734–751
49. Nguyen TT, Pan JS, Chu SC, Roddick JF, Dao TK (2016) Optimization localization in wireless sensor network based on multi-objective firefly algorithm. *JNI* 1(4):130–138
50. Lu S, Zhang N, Qiu Y, Gao Y (2019) A multi-period regret minimization model for uncertain portfolio selection with bankruptcy constraint. *J Intell Fuzzy Syst* 37(6):8417–8439
51. Cheng MY, Prayogo D (2014) Symbiotic organisms search: a new metaheuristic optimization algorithm. *Comput Struct* 139:98–112
52. Chu SC, Du ZG, Pan JS (2020) Symbiotic organism search algorithm with multi-group quantum-behavior communication scheme applied in wireless sensor networks. *Appl Sci* 10(3):930
53. Kumar S, Tejani GG, Mirjalili S (2019) Modified symbiotic organisms search for structural optimization. *Eng Comput* 35(4):1269–1296
54. Tran DH, Cheng MY, Prayogo D (2016) A novel multiple objective symbiotic organisms search (MOSOS) for time–cost–labor utilization tradeoff problem. *Knowl Based Syst* 94:132–145
55. Tejani GG, Pholdee N, Bureerat S, Prayogo D (2018) Multiobjective adaptive symbiotic organisms search for truss optimization problems. *Knowl Based Syst* 161:398–414
56. Abdullahi M, Ngadi MA, Dishing SI, Ahmad BI et al (2019) An efficient symbiotic organisms search algorithm with chaotic optimization strategy for multi-objective task scheduling problems in cloud computing environment. *J Netw Comput Appl* 133:60–74
57. Tejani GG, Pholdee N, Bureerat S, Prayogo D, Gandomi AH (2019) Structural optimization using multi-objective modified adaptive symbiotic organisms search. *Expert Syst Appl* 125:425–441
58. Wolpert DH, Macready WG (1997) No free lunch theorems for optimization. *IEEE Trans Evol Comput* 1(1):67–82
59. Chen W, Wang Y, Zhang J, Lu S (2017) Uncertain portfolio selection with high-order moments. *J Intell Fuzzy Syst* 33(3):1397–1411
60. Markowitz HM, Usmen N (1996) The likelihood of various stock market return distributions, part 2: empirical results. *J Risk Uncertainty* 13:221–247
61. Scott RC, Horvath PA (1980) On the direction of preference for moments of higher order than the variance. *J Finance* 35(4):915–919
62. Sun Q, Yan YX (2003) Skewness persistence with optimal portfolio selection. *J Bank Financ* 27(6):1111–1121
63. Amihud Y (2002) Illiquidity and stock returns: cross-section and time-series effects. *J Financial Mark* 5(1):31–56
64. Yelghi A, Köse C (2018) A modified firefly algorithm for global minimum optimization. *Appl Soft Comput* 62:29–44
65. Zhang J, Teng YF, Chen W (2019) Support vector regression with modified firefly algorithm for stock price forecasting. *Appl Intell* 49(5):1658–1674
66. Devaney R (2018) An introduction to chaotic dynamical systems. CRC Press
67. Deb K, Pratap A, Agarwal S, Meyarivan T (2002) A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Trans Evol Comput* 6(2):182–197
68. Bacanin N, Tuba M (2014) Firefly algorithm for cardinality constrained mean-variance portfolio optimization problem with entropy diversity constraint. *Sci World J* 2014:721521
69. Pouya AR, Solimanpur M, Rezaee MJ (2016) Solving multi-objective portfolio optimization problem using invasive weed optimization. *Swarm Evol Comput* 28:42–57
70. Jin X, Chen N, Yuan Y (2019) Multi-period and tri-objective uncertain portfolio selection model: a behavioral approach. *North Am J Econ Finance* 47:492–504
71. Zhang QF, Li H (2007) MOEA/D: a multiobjective evolutionary algorithm based on decomposition. *IEEE Trans Evol Comput* 11(6):712–731
72. Corne DW, Jerram NR, Knowles JD, Oates MJ (2001) PESA-II: Region-based selection in evolutionary multiobjective optimization. In: *Proceedings of the genetic and evolutionary computation conference (GECCO)*, pp 283–290
73. Mirjalili S, Lewis A (2016) The whale optimization algorithm. *Adv Eng Softw* 95:51–67
74. Askarzadeh A (2016) A novel metaheuristic method for solving constrained engineering optimization problems: crow search algorithm. *Comput Struct* 169:1–12
75. Pan WT (2012) A new fruit fly optimization algorithm: taking the financial distress model as an example. *Knowl Based Syst* 26(26):69–74
76. Van DA, Gary V, Lamont B (1999) Multiobjective evolutionary algorithm research: a history and analysis. *Evol Comput* 8(2):125–147
77. Zitzler E, Deb K, Thiele L (2000) Comparison of multiobjective evolutionary algorithms: Empirical results. *Evol Comput* 8(2):173–195
78. Schott JR (1995) Fault tolerant design using single and multicriteria genetic algorithm optimization. *Cell Immunol* 37:1–13
79. Chen W, Xu HL, Jia L, Gao Y (2020) A machine learning model for Bitcoin exchange rate prediction using economic and technology determinants. *Int J Forecast*. <https://doi.org/10.1016/j.ijforecast.2020.02.008>
80. Jiang M, Jia L, Chen Z, Chen w (2020) The two-stage machine learning ensemble models for stock price prediction by combining mode decomposition, extreme learning machine and improved harmony search algorithm. *Ann Oper Res*. <https://doi.org/10.1007/s10479-020-03690-w>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



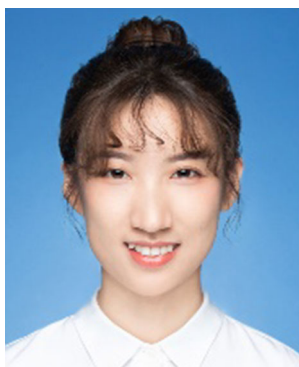
lished 9 textbooks, and 4 patents.

Shan Lu was born in Harbin, China, in 1978. He received the Ph.D. degree in Information from Capital University of Economics and Business, Beijing, China, in 2015. He is currently an Associate and Professor with Capital University of Economics and Business of Management Science and Engineering. He has hosted and participated in more than 20 scientific research projects, published more than 40 research papers, published 2 monographs, pub-



Lifan Jia received the M.S degree from North China Electric Power University, Beijing, China, in 2017, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2019. She is currently an Assistant Professor with School of Management and Engineering, Capital University of Economics and Business, Beijing, China. She has authored or coauthored 22 articles on several SCI journals including Fuzzy Optimization and Decision

Making, Engineering Optimization, Applied Soft Computing, Applied Mathematics and Computation, Soft Computing, Journal of Intelligent Fuzzy Systems, and Journal of Ambient Intelligence and Humanized Computing. Her current research interests include uncertain programming, uncertain differential equations, and their applications.



Ning Zhang was born in Langfang, China, in 1996. She received the bachelor's degree from the school of Computer Science, Harbin Finance University, Harbin, China, in 2018. She is currently a master graduate student majoring in Management Science and Engineering in School of Management and Engineering, Capital University of Economics and Business. Her current research includes portfolio optimization and intelligent algorithm.

Reproduced with permission of copyright owner. Further reproduction
prohibited without permission.